

Commvault Professional

Course Guide

Module 1 – The CommCell Environment

Commvault Platform Release 2023E

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Learning Objectives

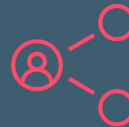
After completing this module, you will:

1



Understand the Commvault Software terminology, with conceptual explanations of CommCell components.

2



Gain insight into the operation of the CommServe server, MediaAgents, servers, and backup storage, along with comprehending data transfer processes like backup, restore, and reporting within a CommCell group.

This course is designed to provide a high-level understanding of Commvault software terminology with conceptual explanations of CommCell components. You will also be able to understand the operation of the CommServe, MediaAgent and backup storage.

Course Modules

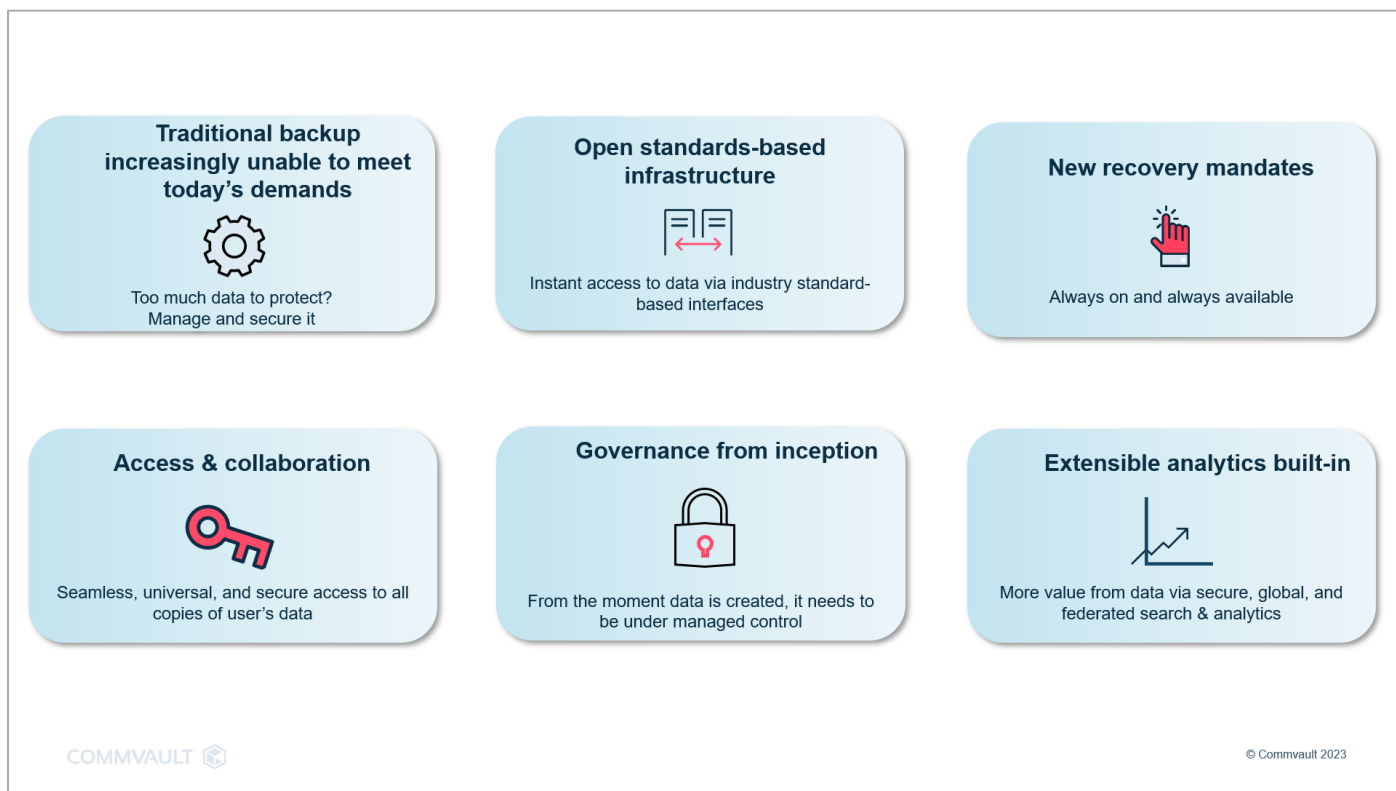
 The CommCell Environment	 Security	 Data Management
 Storage Configuration	 Servers and Features	 Monitoring

COMMVAULT 

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CommVault Overview

Where Commvault Fits in Your Environment



How does the Commvault platform enable your enterprise to be successful in its transformation to a services-oriented architecture? Consider legacy and point products that focus on traditional protection methods such as tape and disk storage with off-site copies for disaster recovery purposes. These solutions may lack complexity, but they fail to align with the overall realities of modern IT infrastructure needs.

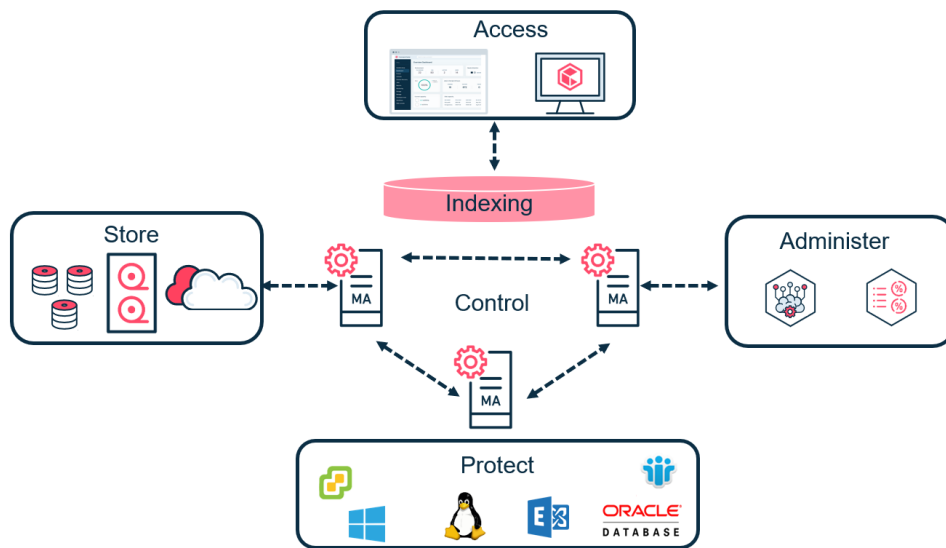
Data management is only as valuable as the effective ability to access the data. High-availability solutions including clustering, replication, snapshots, and virtualization provide added value through quick recovery times and more frequent recovery points. The complexity to manage all these solutions increases without a centralized management system.

Commvault software overcomes the complexity in managing data protection and data access through its platform. The industry's largest supported number of hardware vendors, cloud providers, and virtualization solutions are all managed through Commvault's central platform improving data availability and recovery times.

Data protection and access only tell a portion of the story. The ability to share data across all devices, including mobile Bring Your Own Device (BYOD) devices and endpoint access from any system that uses native applications such as Windows Explorer, Outlook, or standard web browsers, should be ubiquitous for end users. Commvault software provides all these capabilities with intuitive end-user features.

Finally, for many organizations, especially those with hybrid data services with content in the data center, as well as hosted or As a Service, compliance of managed data may be the most complex solution to implement, but it provides the highest value. Intelligent legal holds, records management, analytics, eDiscovery, and data loss prevention are all centrally managed through the Commvault platform.

Commvault Software



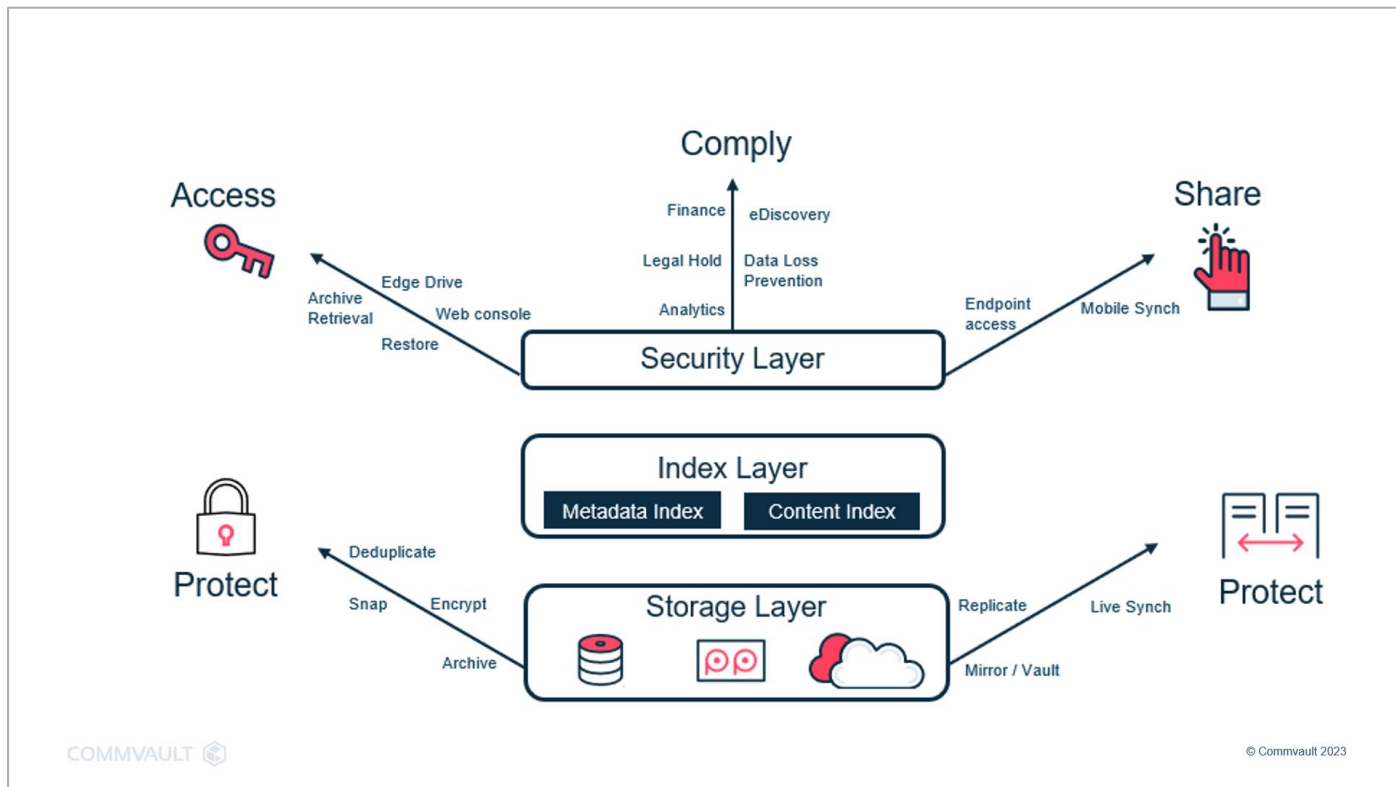
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The Commvault platform provides a fully integrated software suite guaranteeing flexibility in infrastructure, platform, compute, and storage providers.

- **Data Access** – Documents and messages are retrieved seamlessly from mobile devices, web pages, and native application tools.
- **Storage** – Data is protected and stored on disk, tape, and cloud storage locations. Advanced features such as deduplication to disk and tape, job and object retention, and independent multiple copy management are used to meet all Disaster Recovery (DR) and compliance requirements.
- **Indexing** – A distributed self-protecting indexing structure maintains metadata information for all protected data.
- **Data Protection** – File system and application agents are deployed to production systems to protect all major operating systems and applications.
- **Administration** – Most administrative tasks are centrally managed in the Command Center. Tasks can also be performed from mobile devices and web pages.
- **Management** – All activity is centrally controlled and managed using a common code base. Whether using traditional backups, replication, archiving, or snapshots to protect physical or virtual environments, the same core technology is used to provide a solid and flexible management platform.

Commvault Content Store



The Commvault platform structures all managed data in virtual containers using the Content Store. The Content Store is a storage abstraction that allows for data that resides within it to be viewed as a single entity, no matter where that data is otherwise stored.

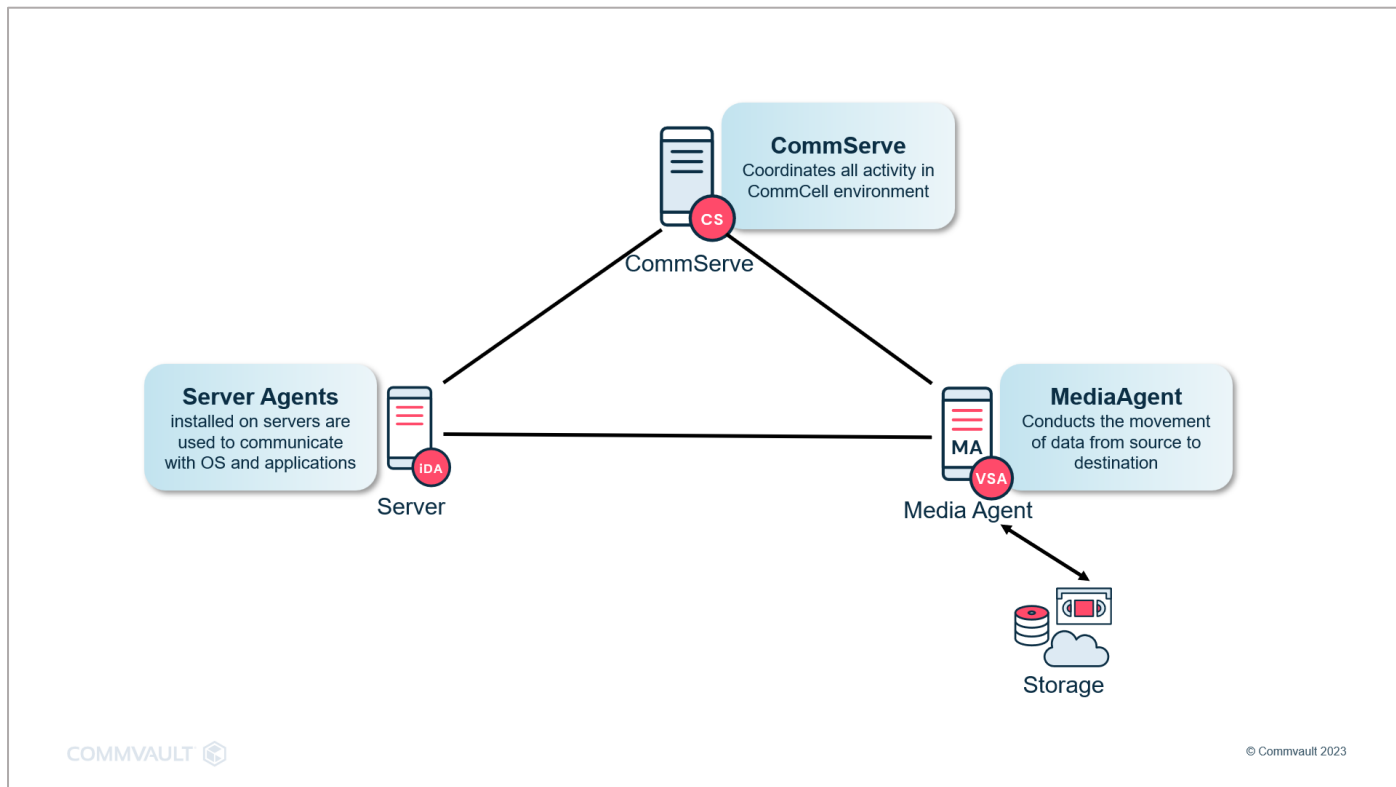
Data in the Content Store is protected and retained in the storage layer. Functionality, including deduplication, archiving, hardware snapshot management, and FIPS certified encryption securely protects data to disk, tape, and cloud storage. Protected data is replicated, mirrored, and synchronized to secondary storage devices for near instant access.

The indexing layer of the Content Store maintains lightweight metadata indices for object management and retrieval, as well as content indices for end user and compliance searches.

All protected data and indices are accessed through the security layer, which features full role-based access control security. Commvault security provides granular access control and distributed security roles, allowing managers to assign permissions to their groups so that users access only what they need.

Access is granted to users to retrieve or restore data using web consoles, mobile devices, and endpoint applications. All data managed in the Content Store is sharable through the security layer and accessible from many secure endpoints.

Physical Architecture



Commvault software is deployed in a cell-like structure called a CommCell environment. One or more cells can be deployed to manage small to enterprise global environments.

The central component of a CommCell environment is the CommServe server which coordinates, manages, and monitors all CommCell activity.

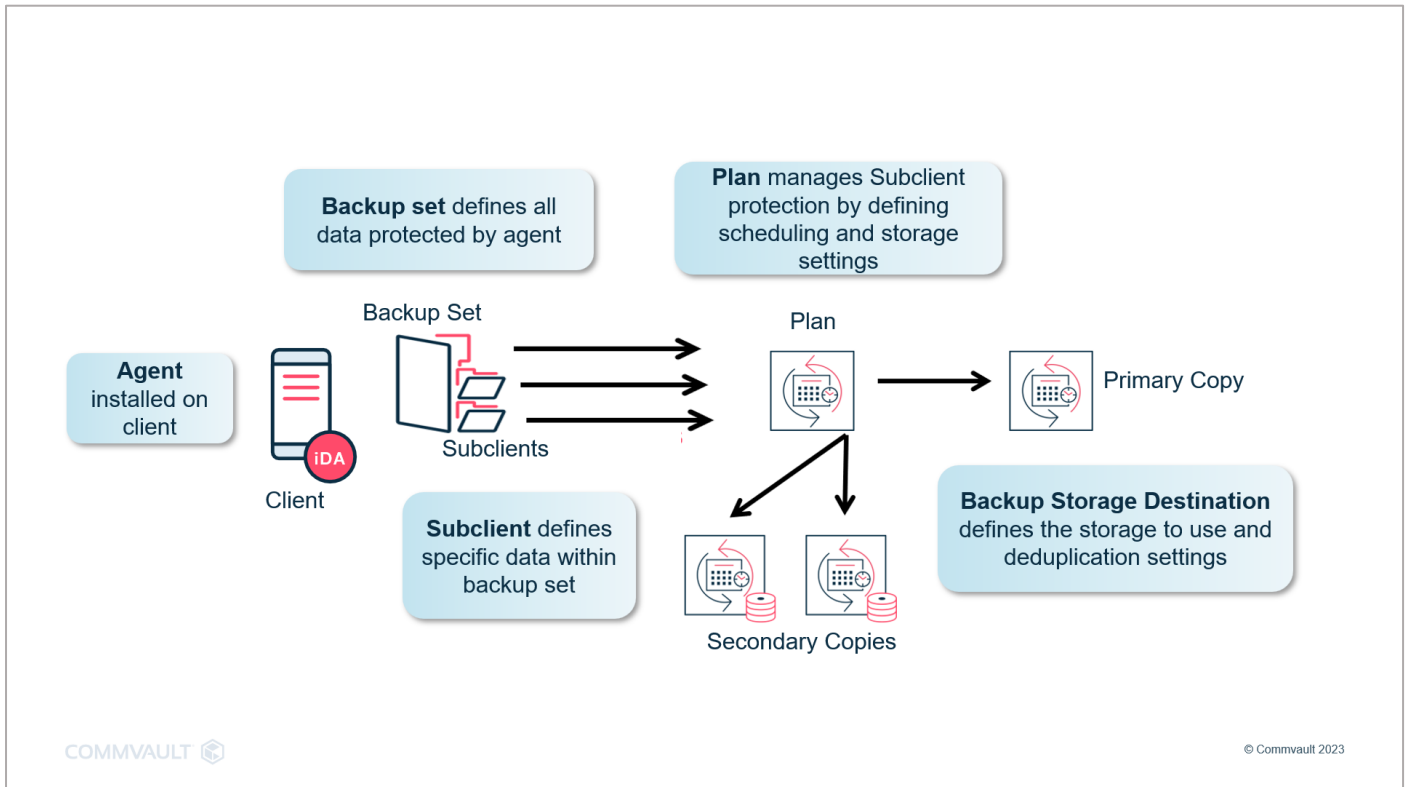
Production data is protected by installing agents which directly communicate with the operating system or application being protected. Any production server with an agent installed is referred to as a client. Data is protected by transferring data through MediaAgents to disk, cloud, or tape storage.

The CommServe server runs on a Windows platform and maintains a Microsoft SQL metadata database. This database contains all critical configuration information. Commvault software does not use a centralized catalog system like most other backup products; thus, the metadata database on the CommServe server will be considerably smaller than databases that contain catalog data.

MediaAgents

A MediaAgent moves data from production systems to protected storage during data protection jobs and moves data back to production systems during recovery operations. It is a software module that can be installed on most operating systems. All tasks are coordinated by the CommServe server. MediaAgents are also used during auxiliary copy jobs when data is copied from a source library to a destination library such as off-site Disaster Recovery (DR) copies.

Logical Architecture



Commvault software logically manages data by containerizing production data, moving the data through logical streams, and managing protected data using plans.

Client Servers and Agents

A client server is any production system that is protected by Commvault software. Client servers use agents to protect the production data by installing the agent directly on the server or using an access node to protect the data. Agents can be deployed in several ways:

- Physical clients can have agents installed directly on them
- Virtual clients can have agents installed directly on them or protected by the Virtual Server Agent (VSA) which would be installed on a physical or virtual access node
- Network Attached Storage (NAS) devices, which cannot have software installed directly on them, are managed and protected by installing NAS agents on access nodes

Backup Sets and Subclients

A backup set is a logical view which contains all data the agent is responsible to protect. Subclients define data that will be protected and how it will be protected. Each subclient container manages specific content within a backup set. Each backup set can have one or more subclients.

Plans

The Commvault software suite offers a wide range of features and options to provide great flexibility in configuring and managing protected data. Protection capabilities such as standard backup, snapshots, archiving and replication can all be incorporated in a single environment for a complete end-to-end data protection solution.

No matter which methods are used within a Commvault environment, the concepts used to manage the data remain consistent.

Planning

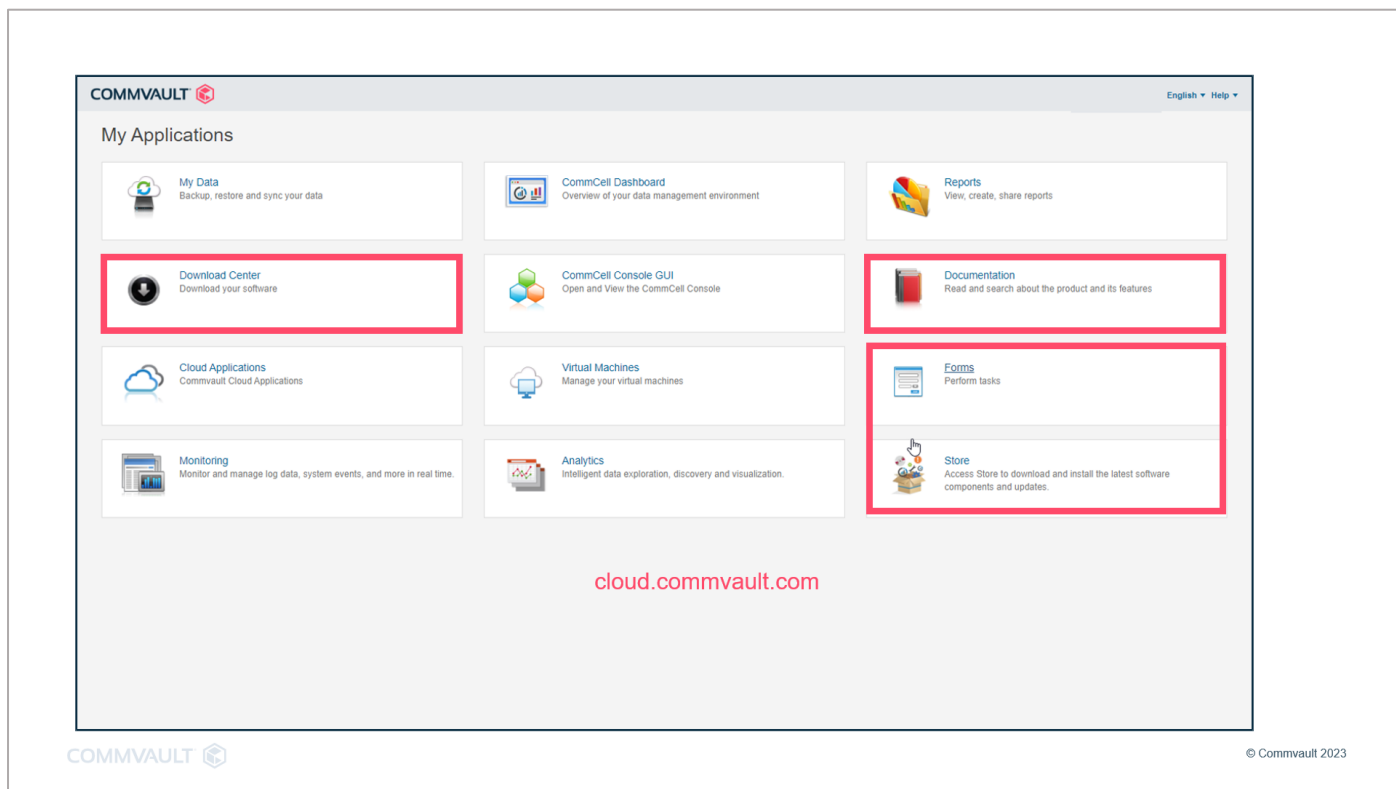
Phases of a Deployment

Installing Commvault software is a straightforward process. However, it is important to note that a full CommCell deployment must be well-planned and executed. Components such as the network, hardware, operating systems, and applications all play a significant role in creating a sound data protection strategy. This section presents the high-level steps and best practices for deploying a CommCell environment.

Deployment Sequence

- Planning
- Download the Commvault software
- Install the CommServe server and executing post-install tasks
- Install the MediaAgents and executing post-install tasks
- Configure storage and deduplication
- Configure solutions
- Apply security

Cloud Resources



The site <http://cloud.commvault.com> provides additional resources to help you manage and maintain a Commvault installation.

- **Download Center** – download feature releases, hot fixes, and binaries for new installations
- **Forms and Store** – download new features and components like workflow or custom reports to install into an existing installation
- **Documentation** – Commvault's technical documentation site

Cloud Resources ...cont'd

The image displays two screenshots of the Commvault user interface. The top screenshot shows the 'My Applications' page, which features a grid of application tiles. The 'Cloud Applications' tile is highlighted with a red rectangle. The bottom screenshot shows the 'Commvault Store' page, which has a dark blue header and a sidebar with navigation links. The main content area is divided into sections for 'Media Kits', 'HyperScale', 'Virtualization', and 'Analytics', each displaying a grid of software components available for download.

My Applications

- My Data**: Backup, restore and sync your data
- Reports**: View, create, share reports
- Download Center**: Download your software
- Cloud Applications**: Commvault Cloud Applications
- Documentation**: Read and search about the product and its features
- Forms**: Perform tasks
- Store**: Access Store to download and install the latest software components and
- Virtual Machines**: Manage your virtual machines
- CommCell Console GUI**: Open and View the CommCell Console

Commvault Store

Stay on top of your data
powered by Datacube

Search

Media Kits

- Commvault Platform Release 2023E
Last updated: Aug 28, 2023
[Download](#)
- Commvault Platform Release 2023
Last updated: Aug 8, 2023
Size: 25.22 MB
[Download](#)
- Miracast
Last updated: Aug 14, 2023
Size: 229.75 MB
[Download](#)
- HyperScale ISO 2.2.212
Last updated: Dec 16, 2022
Size: 5.92 GB
[Download](#)

Virtualization

- Access Node and MediaAgent (FREL) 2023E
Last updated: Jul 7, 2023
Size: 4.15 GB
[Download](#)
- Access Node and MediaAgent (FREL) 2023
Last updated: Feb 14, 2023
Size: 9.12 GB
[Download](#)
- Duplicator Files Dashboard
Last updated: Aug 19, 2022
Size: 74.36 KB
[Download](#)
- Sensitive Files Dashboard
Last updated: Aug 19, 2022
Size: 53.54 KB
[Download](#)

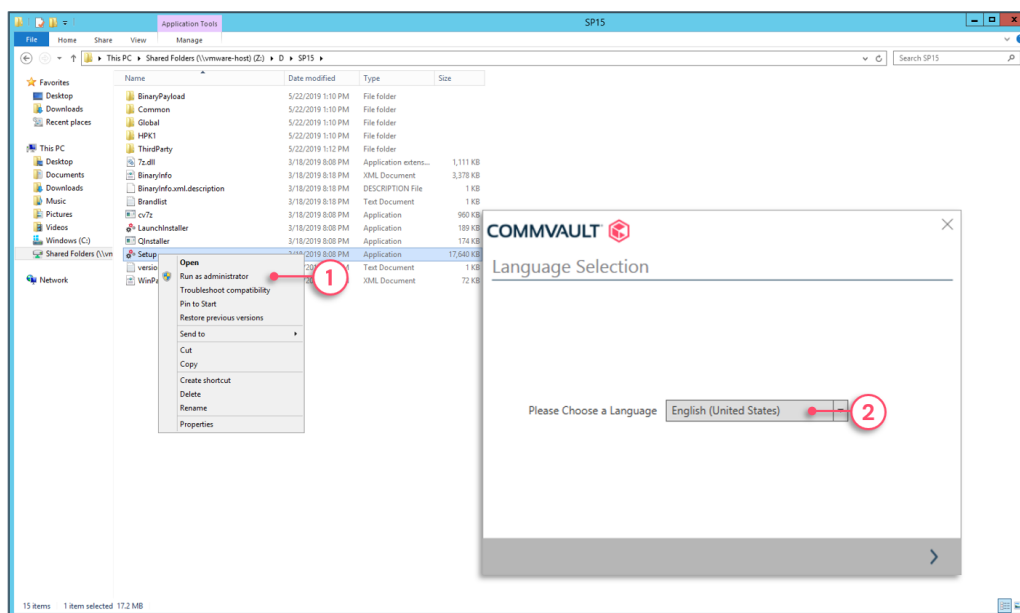
Analytics

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- Click the **Download Center** tile
- Click **Download** for the desired version

CommServe Server Installation

Run the Setup.exe Command



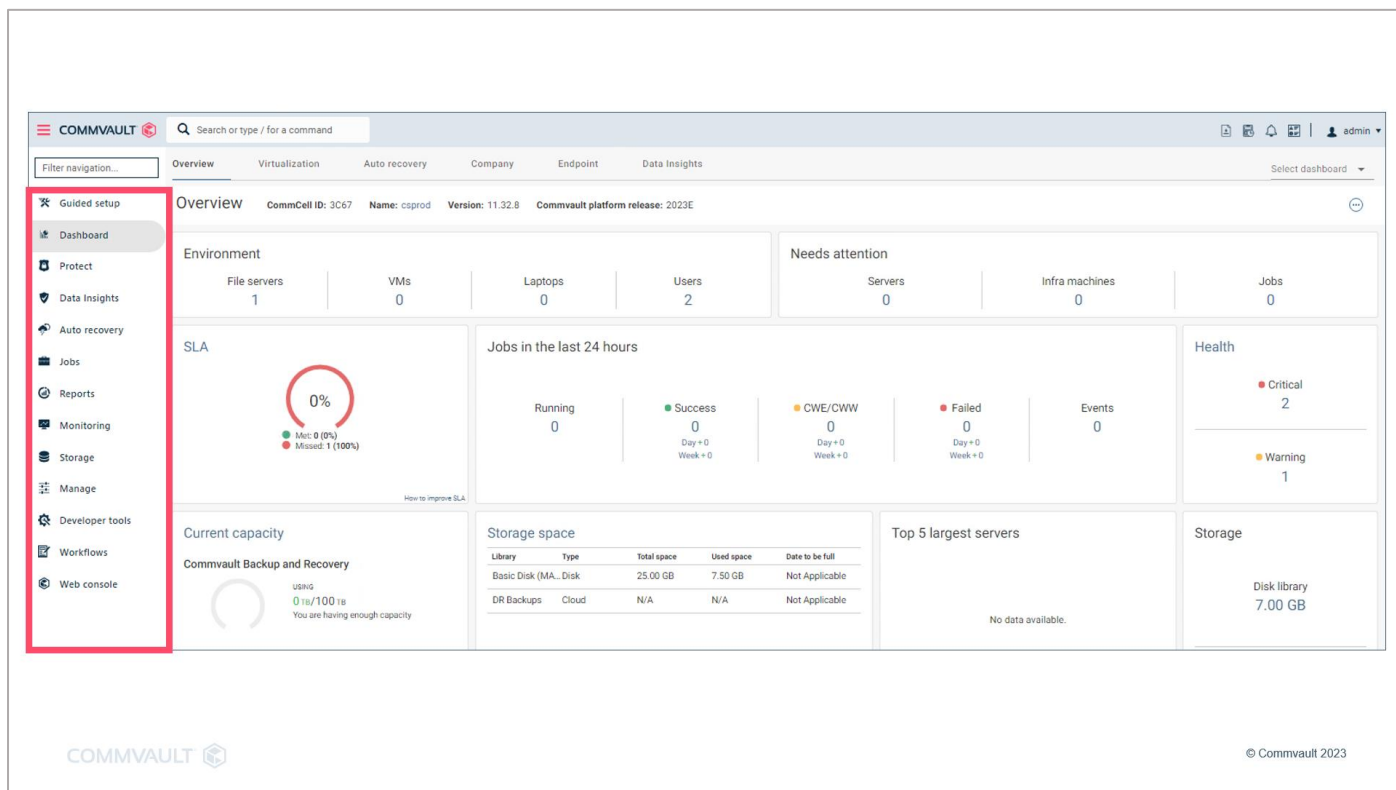
1. Once the file is downloaded, extract the contents into a directory and run the *Setup.exe* as administrator
2. Follow the prompts to install the CommCell, MediaAgent, and optional agents

Deployment Required for CommServe Installation

Attribute	Value (Examples)
Additional Packages	File system agent, MediaAgent, Workflow engine, Web Server
SQL Install Directory	Drive:\Program Files\Microsoft SQL Server
SQL Database Directory	Drive:\Commvault\Database
Commvault Software Directory	Drive:\Program Files\Commvault\ContentStore
DR Share	\\DRCommserve\CSDR
Software Cache	Drive:\Program Files\Commvault\ContentStore\Software

Proper documentation of the CommCell components being installed is essential for a smooth deployment. The following chart provides a sample of the information that must be obtained for the CommServe deployment. Having this information in advance will help the deployment go quicker, bring any shortcomings to the surface, aid in verifying site readiness, and serve as a template for post deployment documentation.

Commvault Command Center Core Setup



The Commvault Command Center is a powerful, easy-to-use web-based user interface for managing on-premises and cloud backup infrastructures. Configure and manage a broad range of features including file systems, databases, and hypervisors backup settings. By using out of the box configurations and streamlined procedures, you can automate common tasks including:

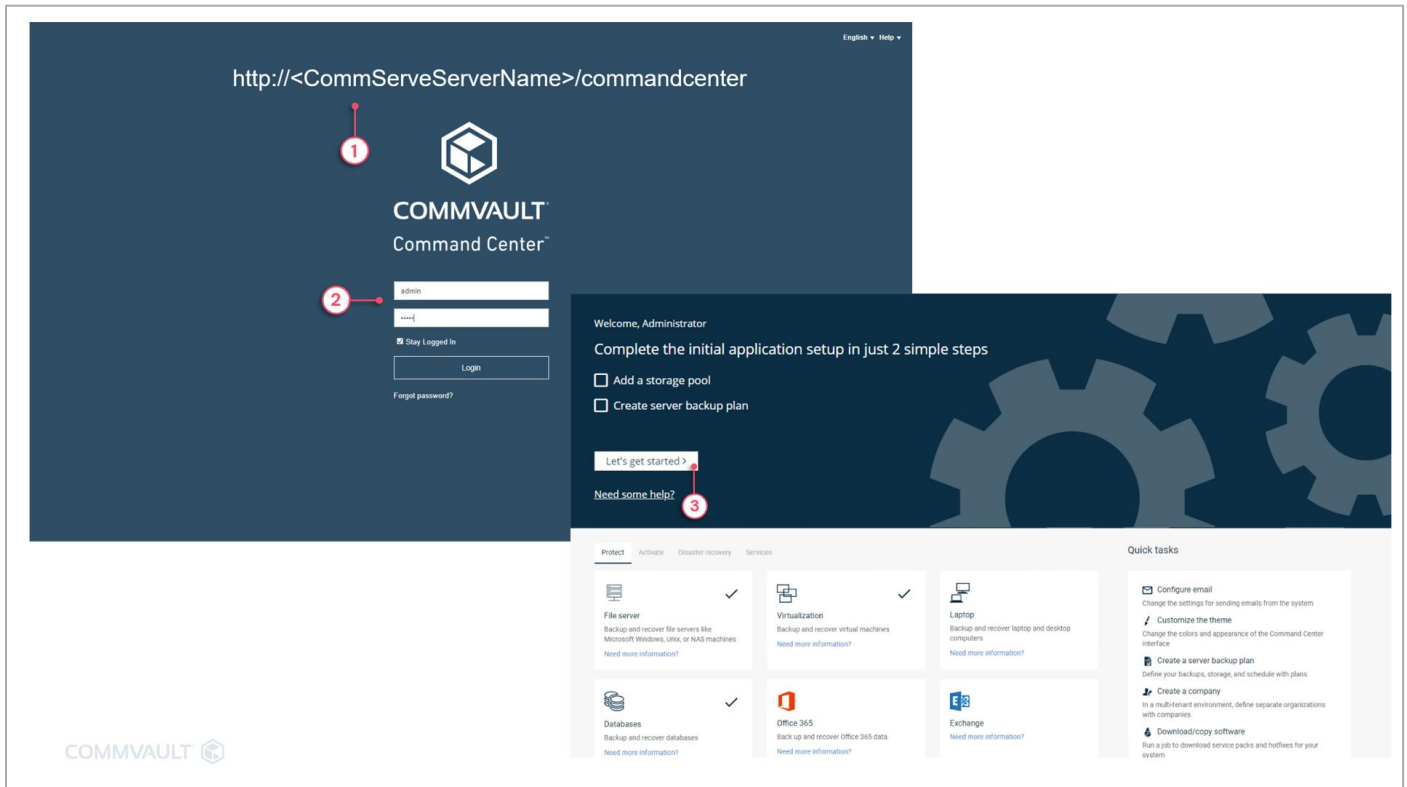
- Setting up your data protection environment features and settings
- Easily configure cloud, disk, and tape storage as backup targets
- Identifying content that you want to protect using Plans
- Setting the RPO for datasets (backup frequency)
- Initiating and monitoring backups and restores

Granular access and permissions are assigned to users, backup administrators, and system administrators providing them with the essential tools required by their respective roles. This level of security safeguards your environment so users and administrators do not modify nor impact environment components that are not relevant to their work.

Managed Service Providers (MSPs)

The Commvault Command Center is used by Managed Service Providers (MSPs) offering backup-as-a-service (BaaS). In a multi-tenant environment, an MSP administrator can mirror real-world organizations by setting up logically separate companies within a single instance of the Commvault software. For each company, the MSP administrator can assign separate Service Level Agreements (SLAs) so that the agreed upon level of data protection is offered to each tenant.

Commvault Command Center Core Setup ...cont'd



1. Open a supported web browser window and enter the URL *http://webhost/commandcenter*
2. Provide your login credentials

Once the Command Center is installed and you are logged in for the first time, the Core Setup wizard is launched from the *Guided setup* section.

3. Click **Let's get started**

Core Setup Sequence

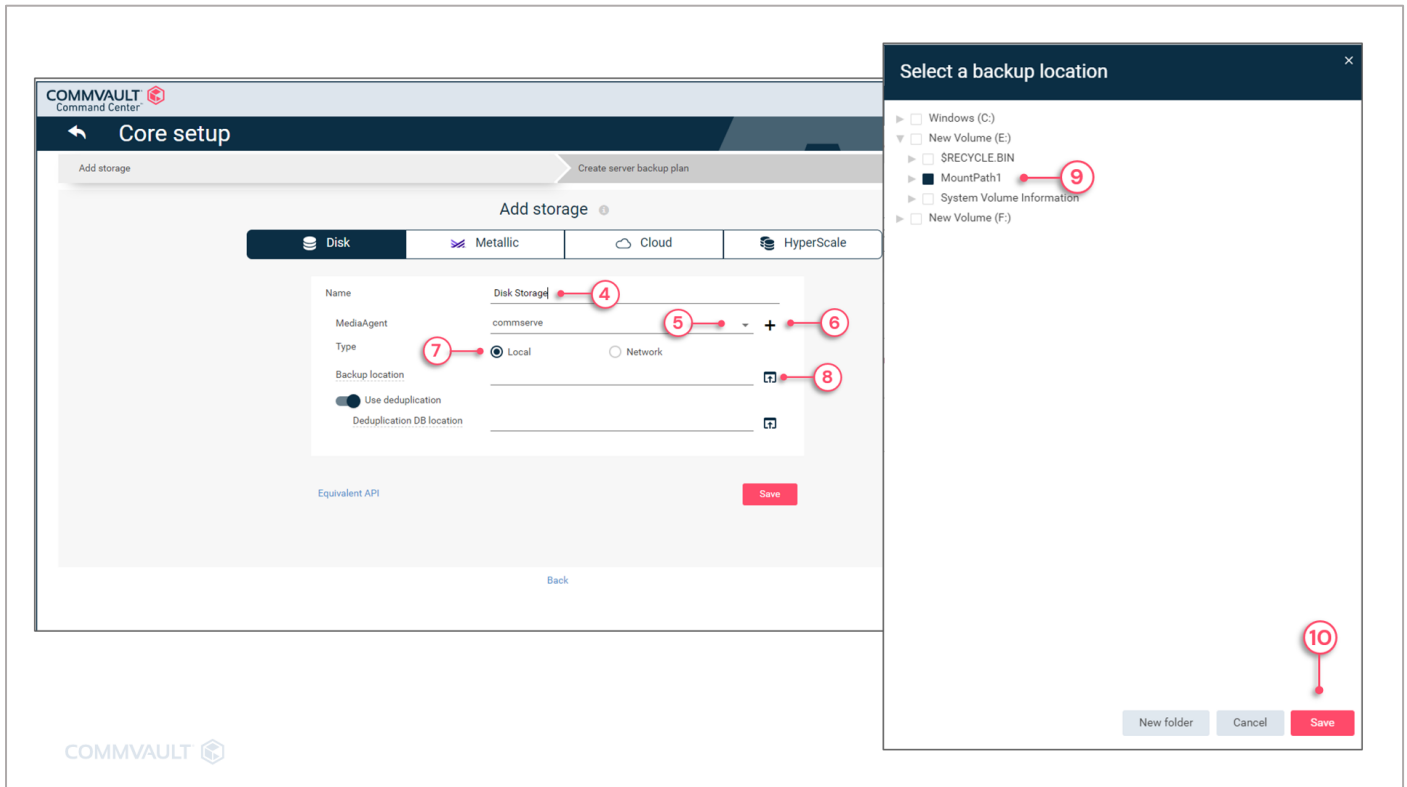
- Add storage
- Configure the first server plan
- Configure an email server

Prepare for Core Setup

- The name and port number of your email server
- The location of the disk library where you want to store backed up data

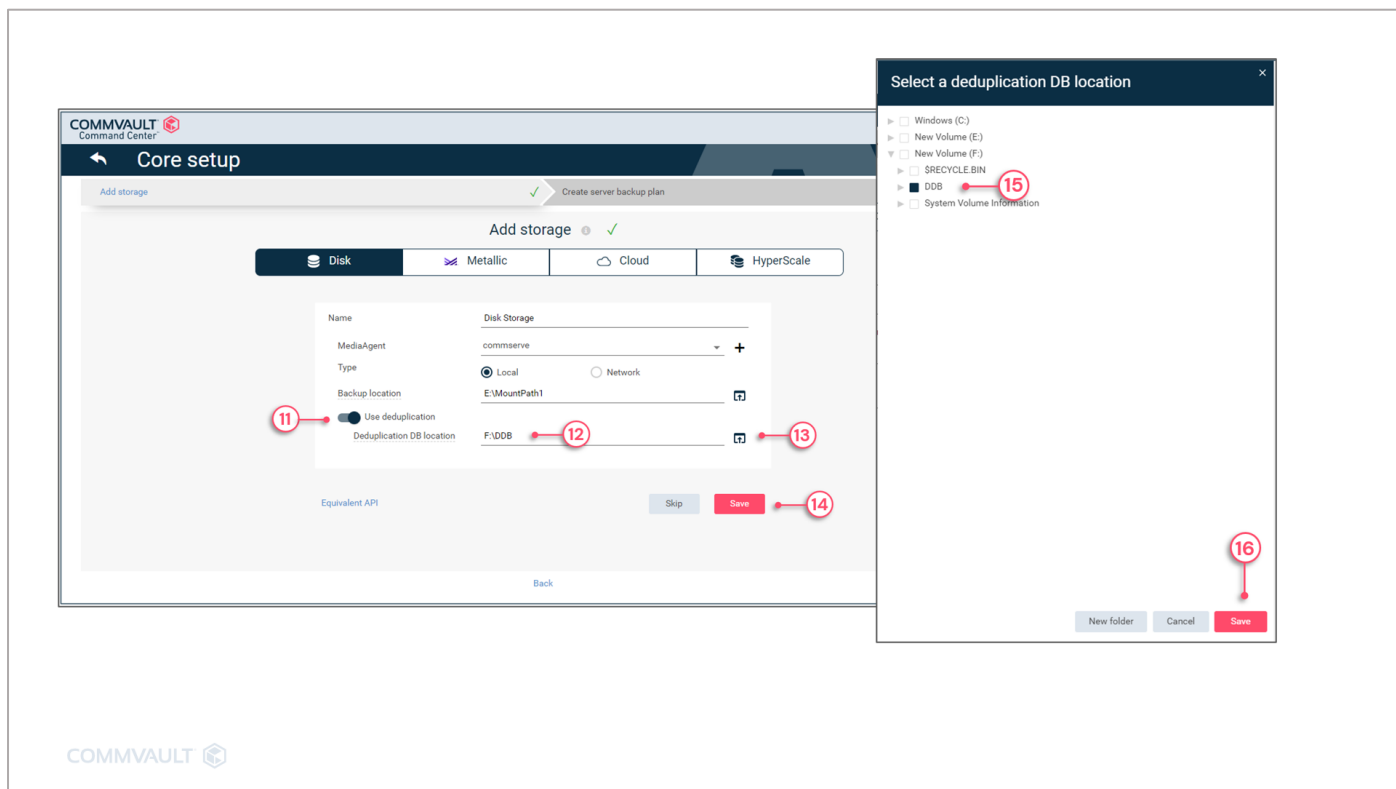
When the Command Center is first launched, a message may display that a permanent license, email, storage, and a backup plan must be configured. Storage can be configured as a local disk, network disk, Metallic cloud storage, or third-party cloud storage.

Commvault Command Center Core Setup ...cont'd



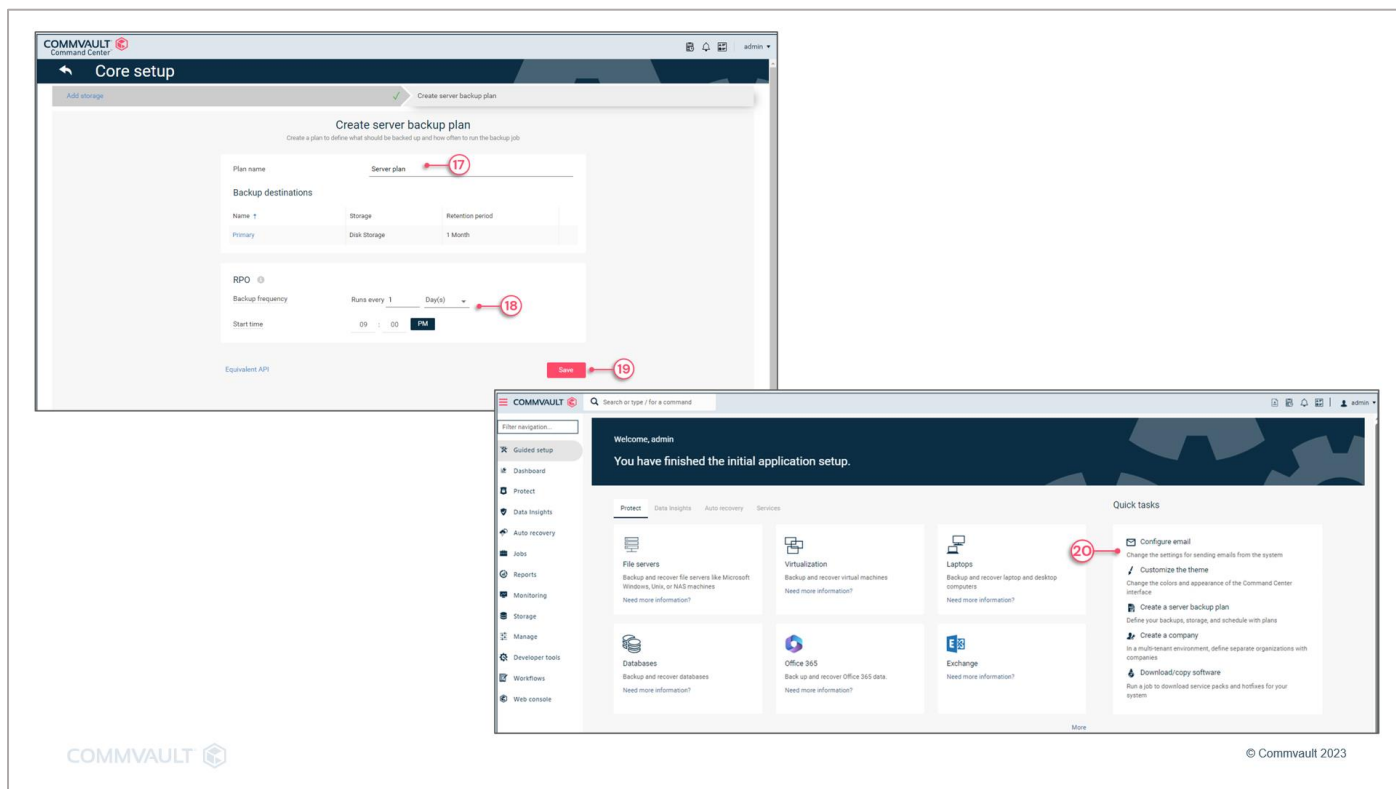
4. Provide a name for the new storage target
5. Use the dropdown list to select the previously installed MediaAgent
6. Click + to install MediaAgent software to a server
7. Select a local or network location for storage
 - a. Select **Local** for direct attached or SAN-based storage
 - b. Select **Network** for NAS storage
8. Click to browse for storage location
9. Check the box to select the mount path for backup location
10. Click **Save**

Commvault Command Center Core Setup ...cont'd



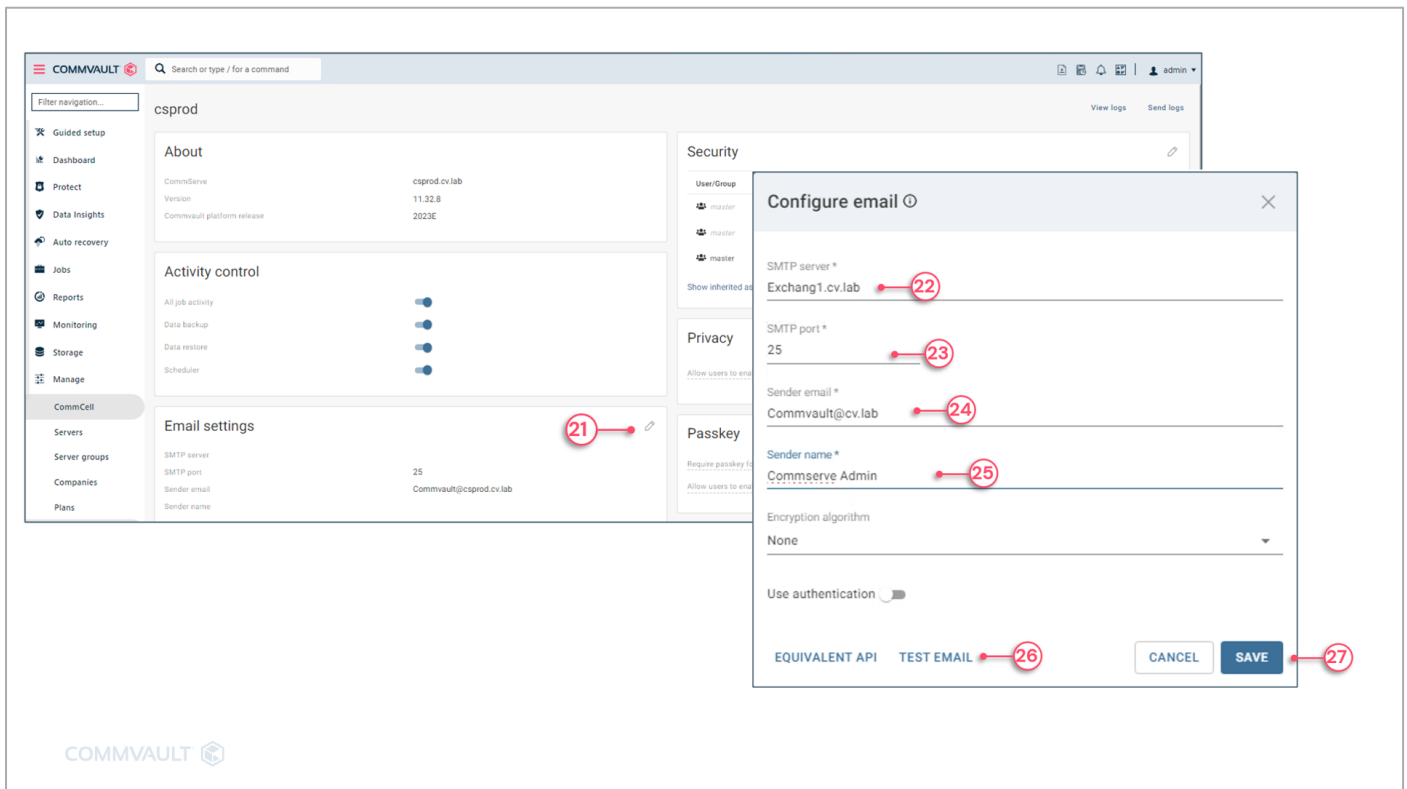
11. Toggle **Use deduplication** on/off to enable/disable deduplication for the storage
12. Enter path to folder for deduplication database
13. Click for option to browser to volume for storing the deduplication database
14. Click **Save**
15. Select the folder to host the DDB partition
16. Click **Save**

Commvault Command Center Core Setup ...cont'd



17. Enter a name for default server plan
18. Define the backup frequency; these settings can be modified later
19. Click **Save**
20. Click **Configure email** to set up connection to the mail server for emailed reports and alerts

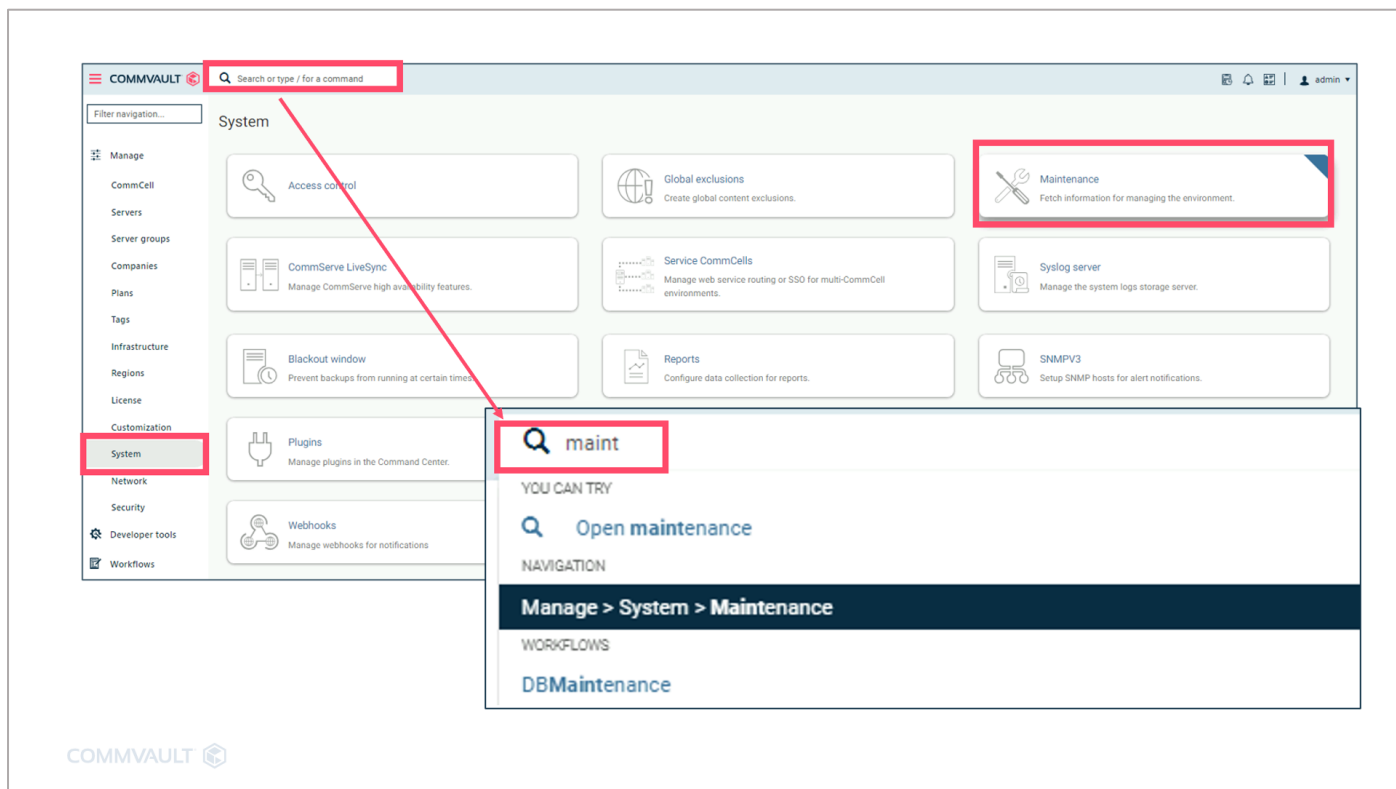
Configure Email



21. From the CommCell Overview page you can also edit the email settings
22. Enter SMTP Email Relay server name
23. Enter SMTP port if different from the default
24. Enter a sender email address for notifications
25. Enter a sender name
26. Optional: Click **Test email** to verify connectivity to SMTP server
27. Click **Save**

Once the email server, storage locations, and backup schedules are set the administrator will see the Overview Dashboard when they first log into the Commvault Command Center.

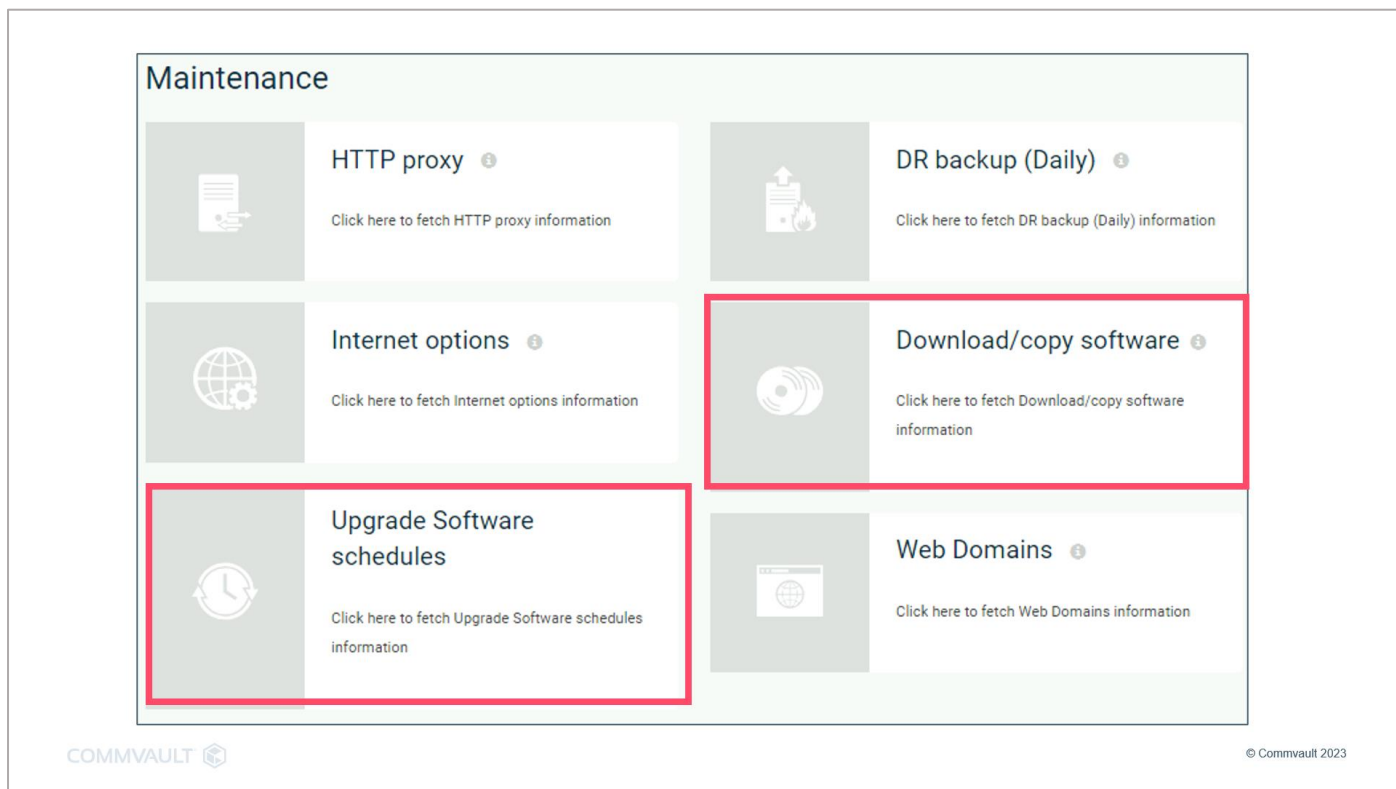
Configure Automatic Updates



Automatic Updates can be configured from the *Navigation pane > System > Maintenance tile*. This can also be found by typing *maintenance* in the search bar.

Once you begin entering the search term, the search bar will start displaying all object and configuration items matching the term.

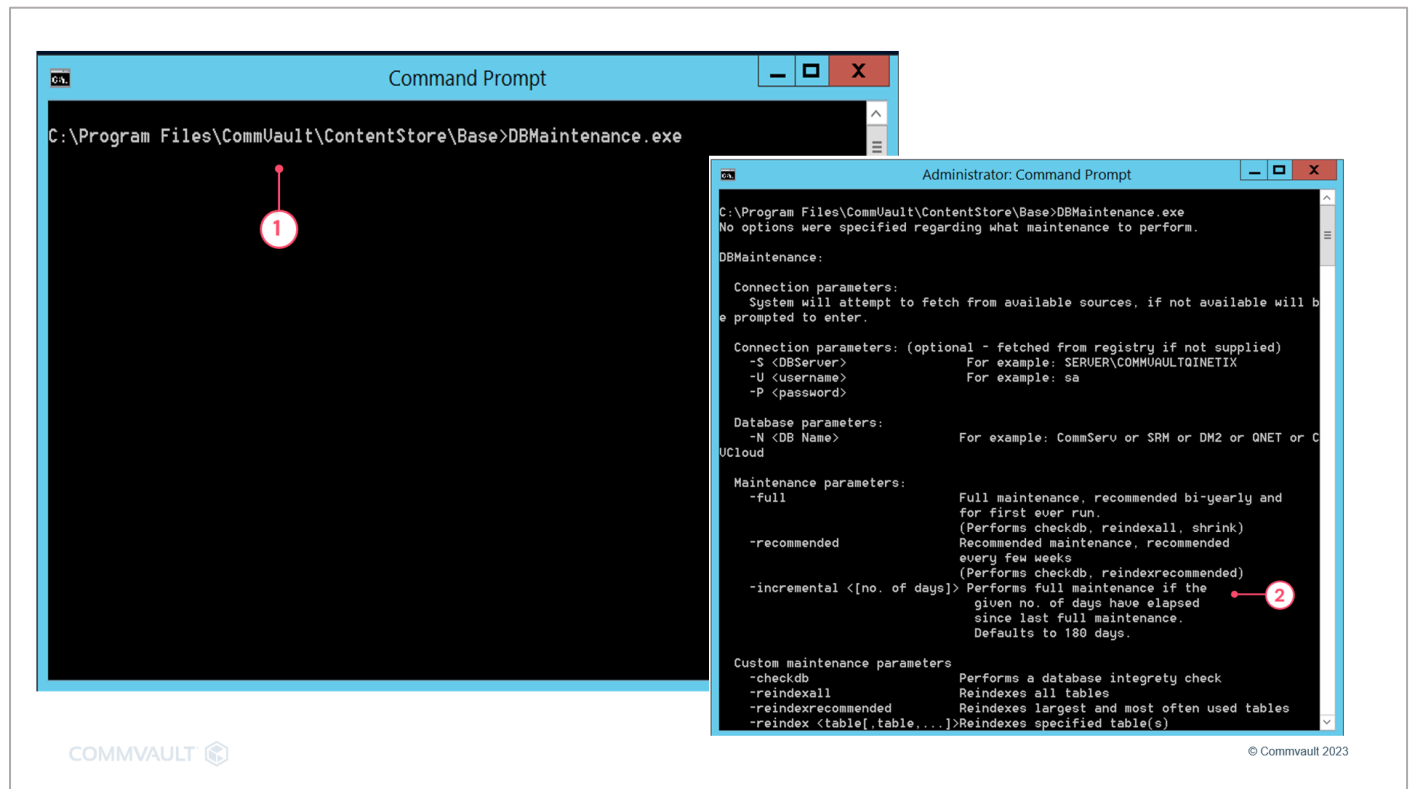
Configure Automatic Updates ...cont'd



Within the maintenance page, the *Download/copy software tile* can be used to populate the software cache with the required software update packages.

The *Upgrade Software schedules tile* allows you to configure the schedules for automatically performing software updates.

CommServe Database Maintenance



CommServe Server Database Maintenance

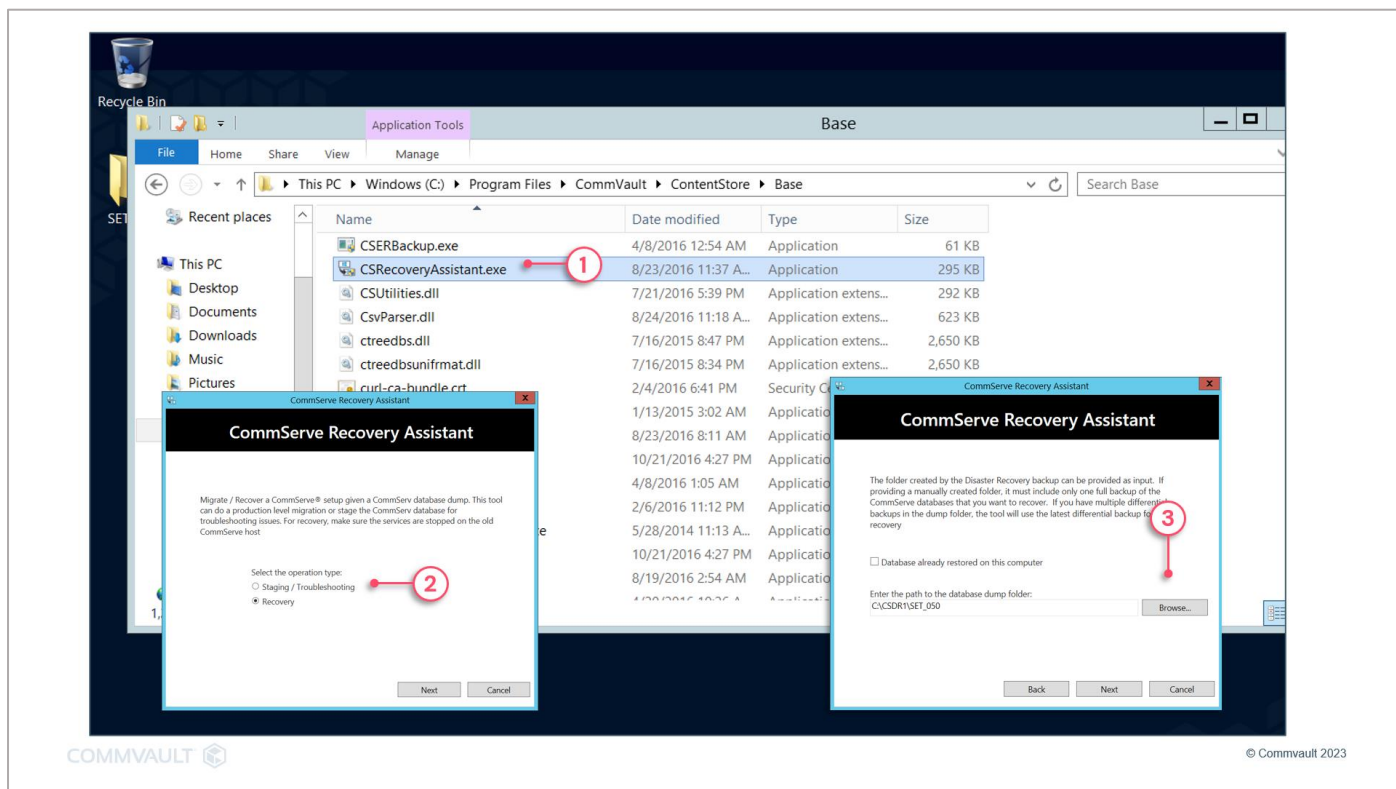
It is crucial to ensure that the CommServe server is maintained for maximum productivity.

1. Access DBMaintenance.exe to optimize the CommServe server database
2. It provides the correct syntax and options for the tool, such as the maintenance mode

Maintenance Modes

- **Full** – Performs a full maintenance on the database. It includes CheckDB, ReindexAll, and ShrinkDB commands, recommended to run this twice each year
- **Recommended** – performs a recommended maintenance including ShrinkDB and ReindexRecommended commands, recommended to run every two weeks; by default, a system created schedule will execute it alternating Sundays
- **CheckDB** – validates the consistency of the CommServe database by running an integrity check
- **ReindexRecommended** – re-indexes the largest and most frequently used tables of the database
- **ReindexAll** – re-indexes all tables of the database
- **ShrinkDB** – if table re-indexing creates a significant amount of fragmentation, the ShrinkDB command will reclaim that space by shrinking the database

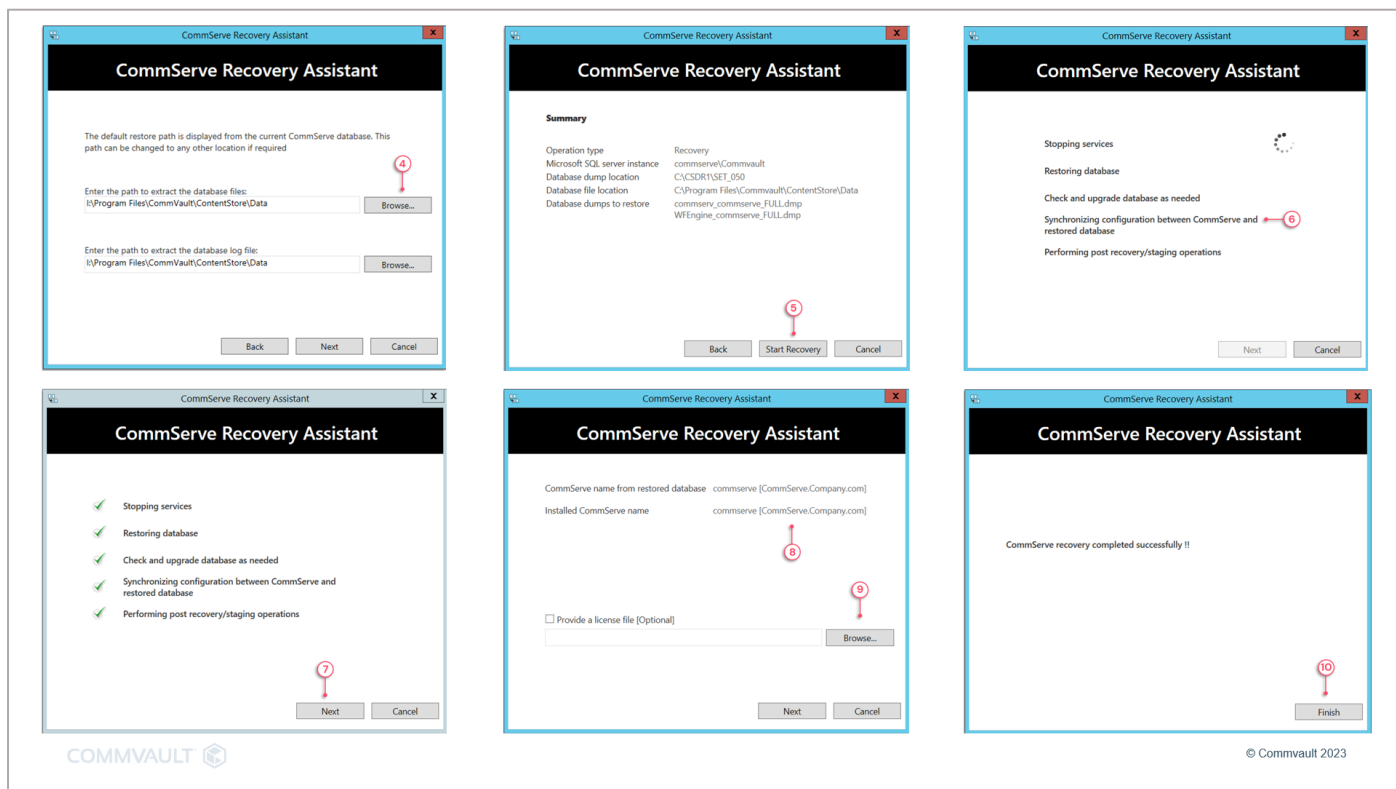
CommServe Recovery Assistant Tool



The CommServe Recovery Assistant tool is used to restore the CommServe database from the DR backup. The tool is used to rebuild the CommServe server on the same or different computer, change the name of the CommServe host, and update the CommCell license.

1. From the installation directory, execute **CSRecoveryAssistant.exe**
2. Choose the operation type
3. Click **Browse** to provide the location of the DR Backup

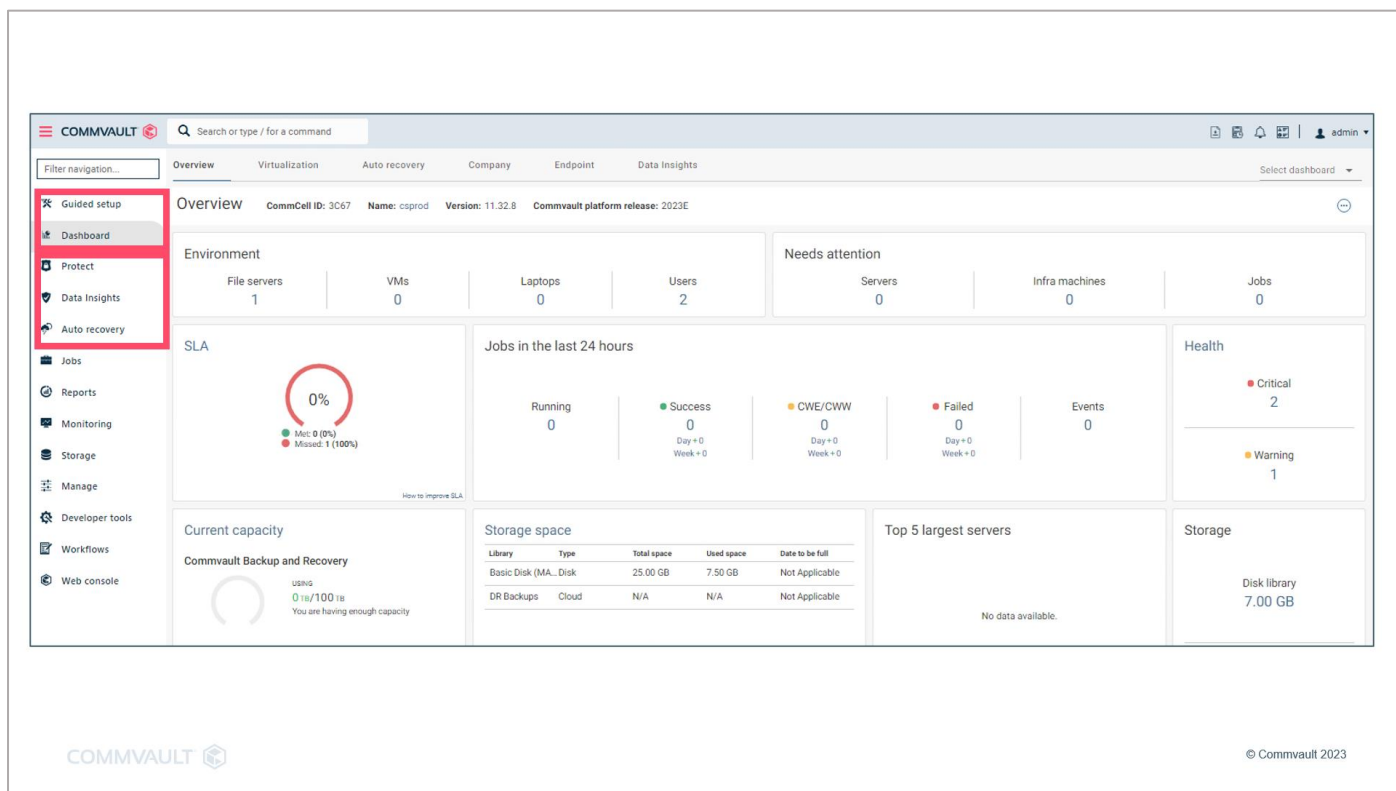
CommServe Recovery Assistant Tool ...cont'd



4. Click **Browse** to provide the location where to restore the database and logs
5. Click **Start Recovery** to start the recovery
6. Status of recovery steps are displayed
7. Once recovery is completed, click **Next**
8. If restoring to an alternate server, the CommServe server name is changed
9. A license file can be provided
10. Once completed, click **Finish**

Commvault Command Center Navigation

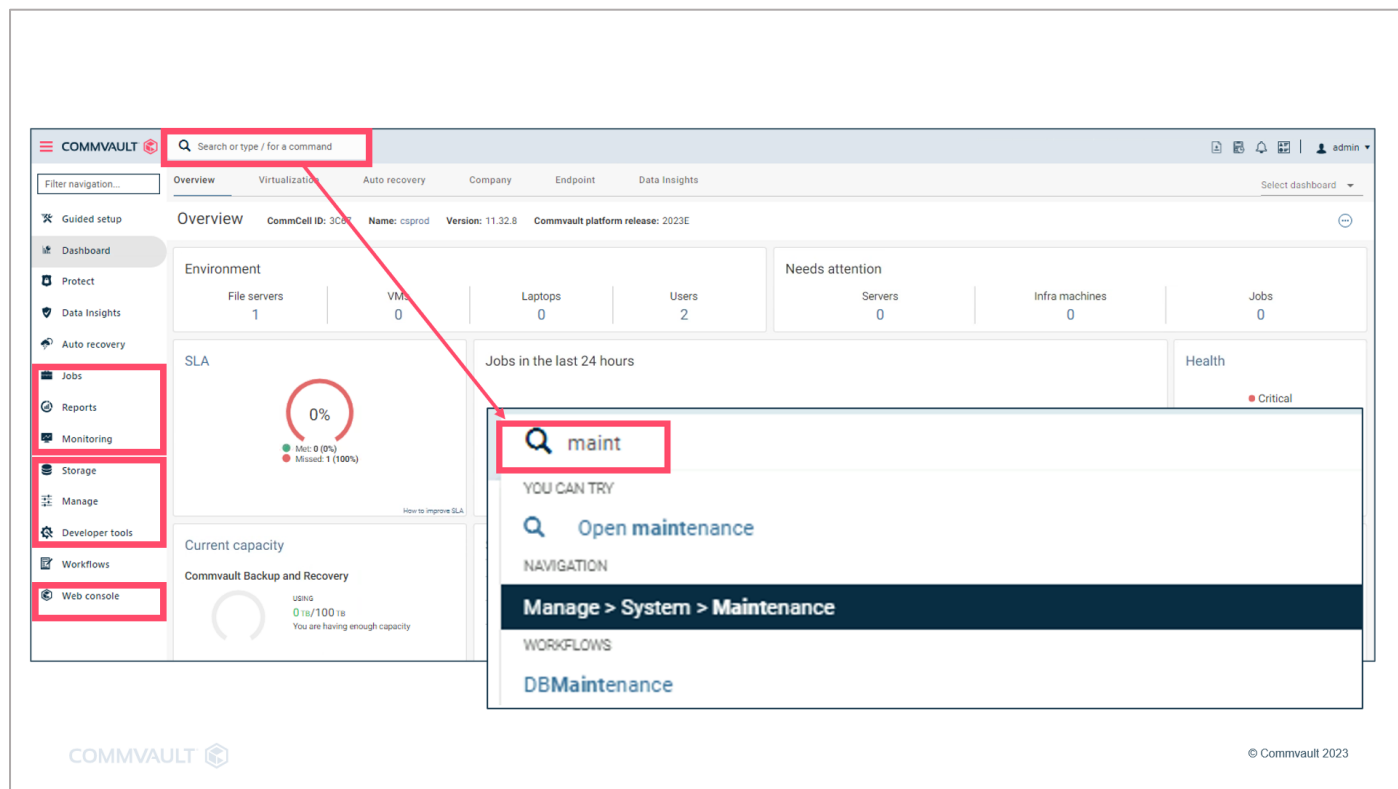
Navigation Overview



The Commvault Command Center navigation is divided into sections. The top section is the *Guided setup* and *Dashboards*. This is where you will manage your initial configuration and is what you see when you first log into the system.

The *Protect*, *Data Insights*, and *Auto Recovery* sections provide views of systems in the data center and in the cloud, based on how they are protected.

Navigation Overview ...cont'd



The *Jobs*, *Reports*, and *Monitoring* sections are day-to-day operational functions that you will use to monitor the health of your environment. The dashboard is a good place to get an overview of daily system health. Click an item in the dashboard for a more detailed view.

The *Storage*, *Manage*, *Developer tools*, and the *Web console* are system-wide configuration and management functions. Typically, these are set up once and then only reviewed or reconfigured as needed. The *Manage* section includes security and system administration. Use these tools to reconfigure your environment when your business needs change.

The Command Center search bar is a very powerful tool to allow users to find anything within the Command Center.

Start typing the element you are looking for and the search tool will provide links to matching objects and configuration elements.

Sidebar Menu

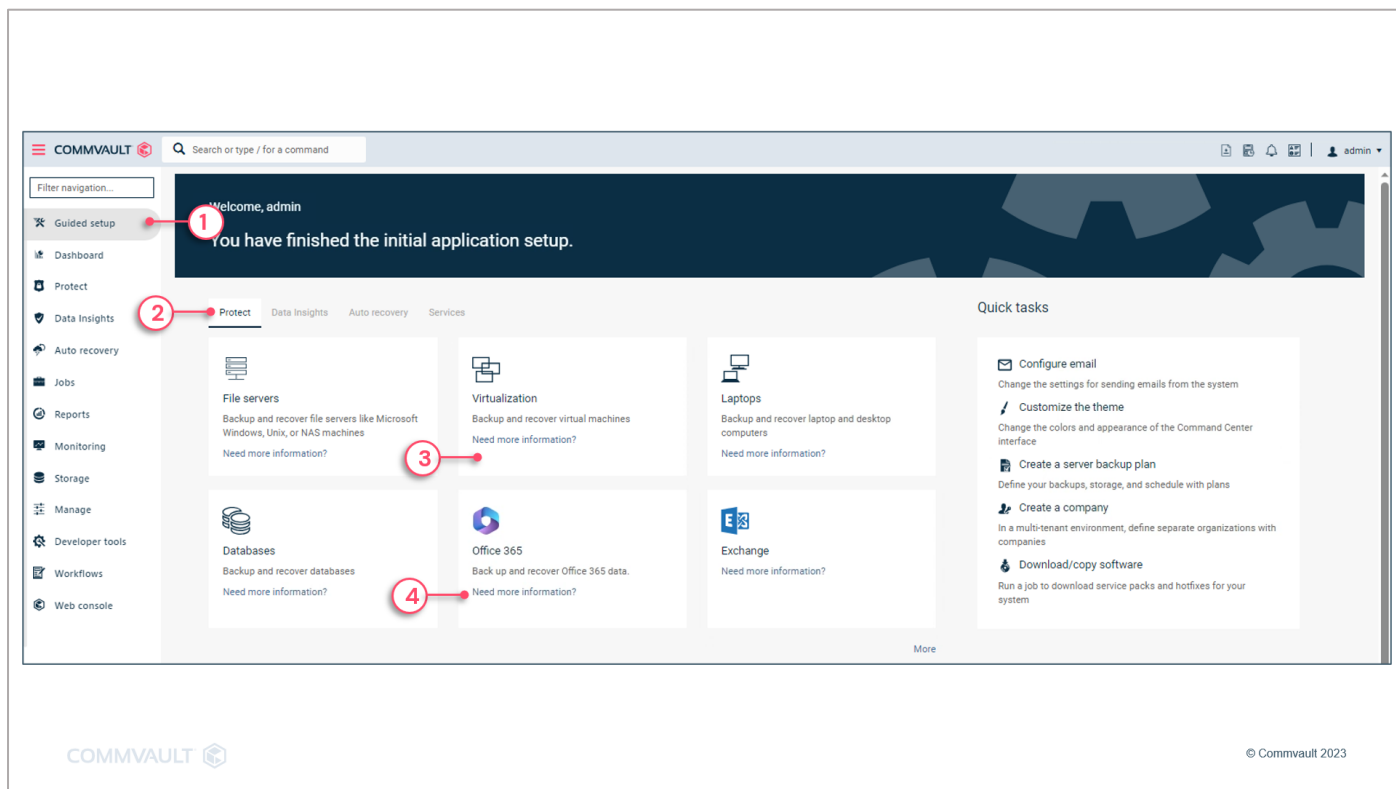
The screenshot shows the Commvault Overview dashboard. A red circle with the number '1' points to the left sidebar menu, which includes options like Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, Storage, Manage, Developer tools, Workflows, and Web console. Another red circle with the number '2' points to the 'Filter navigation...' dropdown at the top of the sidebar. The main dashboard area displays various metrics and charts, including Environment (File servers: 1, VMs: 0, Laptops: 0, Users: 2), Needs attention (Servers: 0, Infra machines: 0, Jobs: 0), Jobs in the last 24 hours (Running: 0, Success: 0, CWE/CWW: 0, Failed: 0, Events: 0), Health (Critical: 2, Warning: 1), Current capacity (0% used), Storage space (Basic Disk: 25.00 GB, Used: 7.50 GB), Top 5 largest servers (No data available), and Storage (Disk library: 7.00 GB).

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1. The left sidebar is the primary menu
2. Use the filter to find a specific selection and use the search bar to find specific entities

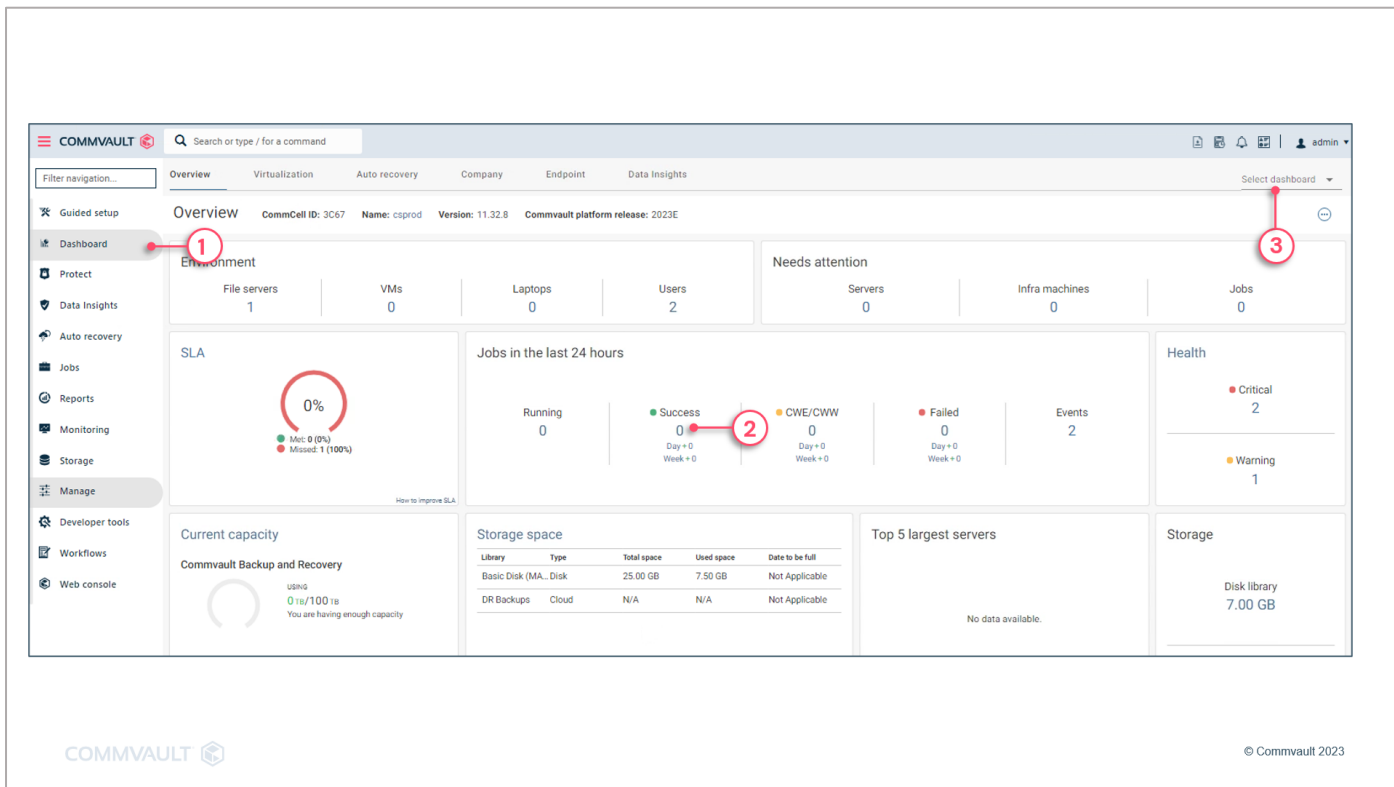
Guided Setup



Use the Guided Setup to configure or add features to your environment.

1. Select *Guided setup* to access the configuration screen
2. Select the **Protect** tab
3. Click a feature tile to add it to the Commvault Command Center
4. Click the link to open Commvault technical documentation for additional information

Dashboard



Daily monitoring with the dashboard is essential to keep your CommCell environment running efficient. It shows key information for a component or aggregation of components. The dashboard provides an overall status view of the CommCell environment and client protection jobs. View information for all clients and features. More views are available in other dashboards.

TIP!

Use your mouse to hover over and identify active areas of the dashboard that can be expanded for additional information.

The dashboard view provides an overall view of the status of the CommCell and client protection jobs. By default, the view displays information about all clients. If needed, other specific dashboards can be accessed.

1. Select **Dashboard** to access the Overview Dashboard for summary information about protected servers, storage capacity, and health in the CommCell environment
2. Active links in the dashboard sections can be expanded with a mouse click to provide additional information
3. Click **Select Dashboard** to access the dropdown list for other dashboards

Jobs

The image displays two screenshots of the Commvault Jobs interface. The top screenshot shows the 'Active jobs' view, which includes a table of running jobs and a progress bar. The bottom screenshot shows the 'Job history' view, which includes a table of past jobs and their status. Red circles with numbers 1 through 6 highlight key UI elements.

Active jobs view (top screenshot):

- 1. Select **Jobs** in the left navigation pane.
- 2. Check boxes to select one or more jobs.
- 3. Choose to kill, suspend, or resume the jobs (Actions column).
- 4. The progress bar displays the progress of the job.

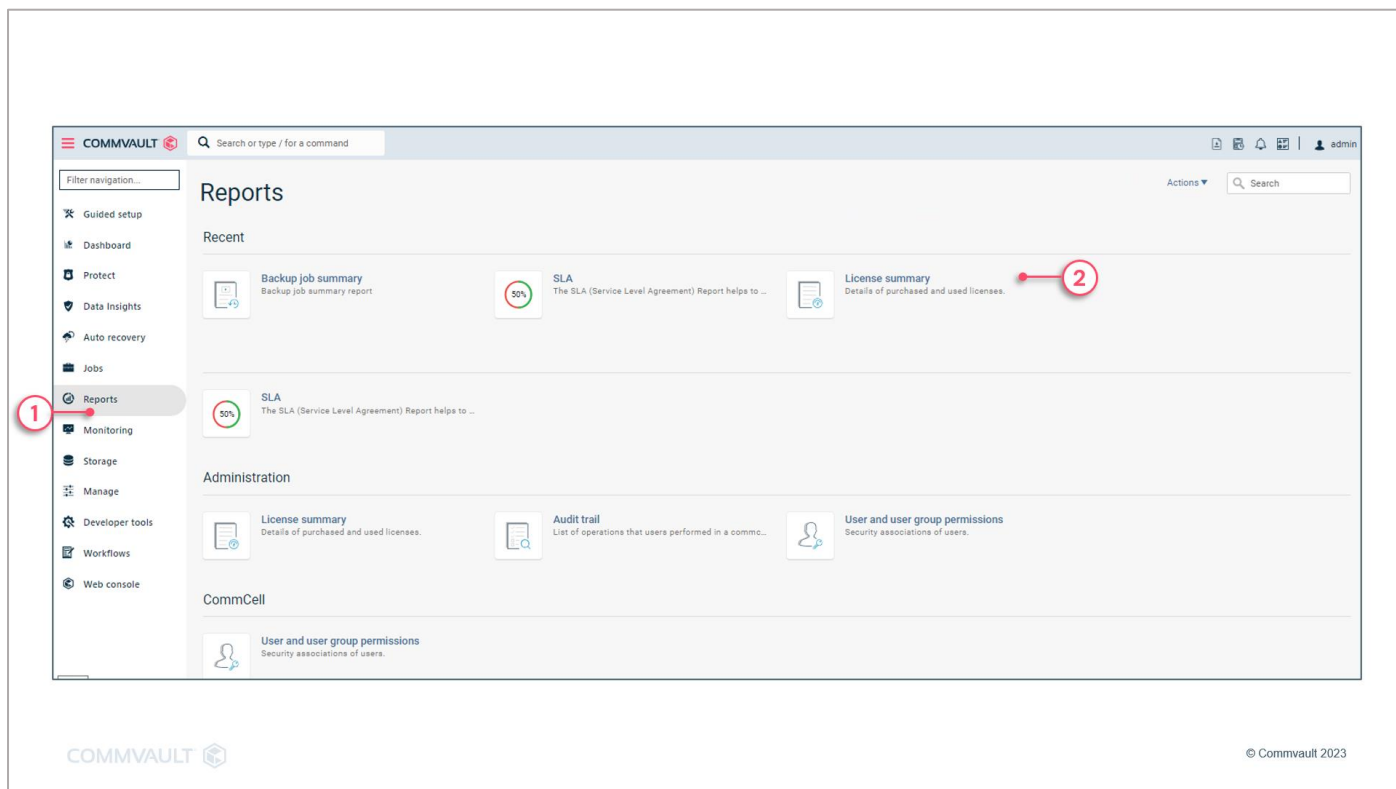
Job history view (bottom screenshot):

- 5. Click **Job history** in the top navigation bar.
- 6. Select the time range from the dropdown list.

The Job view provides monitoring functionality for all active jobs within the CommCell environment. The Jobs view displays current running jobs by default. It can also display the job history range of the last 24 hours or the last three months. Use this view to kill, suspend, and resume jobs, and to view the job logs.

1. Select **Jobs** to see all active jobs
2. Check boxes to select one or more jobs
3. Choose to kill, suspend, or resume the jobs
4. The progress bar displays the progress of the job
5. Click **Job history** to access job history
6. Select the time range from the dropdown list

Reports



The *Reports* view can be expanded to display available reports as tiles. Most reports have a graphic view followed by a detailed table. Apply filters and customize the views. Set up an email schedule to stakeholders.

1. Displays lists of reports to run on-demand and schedule emailed reports to be sent
2. Click a tile to generate its report

The screenshot displays the Commvault Alerts page. On the left, a navigation sidebar includes sections like Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, Security IQ, Alerts (highlighted), Events, Unusual file activity, Approvals, Storage, and Manage. The main area is titled 'Triggered alerts' and contains a table of alerts. A dropdown menu is open for the 'View' column, showing options: All, Critical (last 24 hours), Critical, Major, and Information. The table columns are: Alert info, Type, Detected time, Client name, and Pin. The table lists various alerts such as 'Insufficient storage on library', 'Mount Path went Offline', and 'Job Failed'.

Alert info	Type	Detected time	Client name	Pin
Insufficient storage on library	Media Management - Library Management	Sep 7, 5:31:14 AM		
Insufficient storage on library	Media Management - Library Management	Sep 7, 5:31:14 AM		
Device went offline	Media Management - Device Status	Sep 5, 5:51:30 PM		
Device went offline	Media Management - Device Status	Sep 5, 5:51:30 PM		
Device went offline	Media Management - Device Status	Sep 5, 3:18:55 PM		
Device went offline	Media Management - Device Status	Sep 5, 3:18:55 PM		
Insufficient storage on library	Media Management - Library Management	Sep 5, 3:18:54 PM		
Data aging job failed	Job Management - Data Aging	Jul 26, 12:00:32 PM		
Device went offline	Media Management - Device Status	Jul 26, 11:49:38 AM		
Device went offline	Media Management - Device Status	Jul 26, 11:49:38 AM		
Insufficient storage on library	Media Management - Library Management	Jul 26, 11:19:09 AM		
Data aging job failed	Job Management - Data Aging	Apr 28, 12:00:28 PM		
Data aging job failed	Job Management - Data Aging	Apr 28, 11:09:53 AM		
Insufficient storage on library	Media Management - Library Management	Apr 26, 5:06:09 PM		
Device went offline	Media Management - Device Status	Apr 26, 5:06:09 PM		

1. Displays all configured alerts in the CommCell environment
2. Filter using the dropdown list
3. Use *Search* to locate Alerts by content

Events

The screenshot displays the Commvault Events view. The top navigation bar includes the Commvault logo, a search bar, and user information. The left sidebar contains a navigation menu with options like Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, Security IQ, Alerts, Events, Unusual file activity, Approvals, Storage, Manage, and Developer tools. The main content area shows a list of events with columns for Event ID, Severity, Job ID, Date, Program, Computer, Event code, and Description. A search bar is located at the top right of the event list. Below the event list, a detailed view for event 1666 is shown, including fields for Severity, Date, Event code, Program, Computer, Job ID, and a Description box.

Event ID	Severity	Job ID	Date	Program	Computer	Event code	Description
1845	Info	41	Sep 7, 7:54 AM	JobManager	csprod	19:1076	Restore job [41] completed. Client [ypserv1]. Agent Type
1820	Info	40	Sep 7, 7:44 AM	JobManager	csprod	19:1076	Restore job [40] completed. Client [ypserv1_1]. Agent Ty
1668	Info		Sep 7, 7:14 AM	MediaManager	csprod	32:43	MediaAgent [ypserv1] is ready.
1667	Info		Sep 7, 7:14 AM	MediaManager	csprod	32:44	MediaAgent [ypserv1] is offline.
1666	Info	37	Sep 7, 7:14 AM	DistributeSoftware	csprod	68:92	Job Completed Successfully
1665	Info	37	Sep 7, 7:13 AM	DistributeSoftware	csprod	68:83	All packages were successfully installed on client [ypserv1]
1664	Info						
1663	Info						
1662	Info						
1661	Info						
1660	Info						
1659	Info						
1652	Info						

Event details for event 1666:

Event		Source	
Severity	Info	Program	DistributeSoftware
Date	Sep 7, 7:14 AM	Computer	csprod
Event code	68:92	Job ID	37

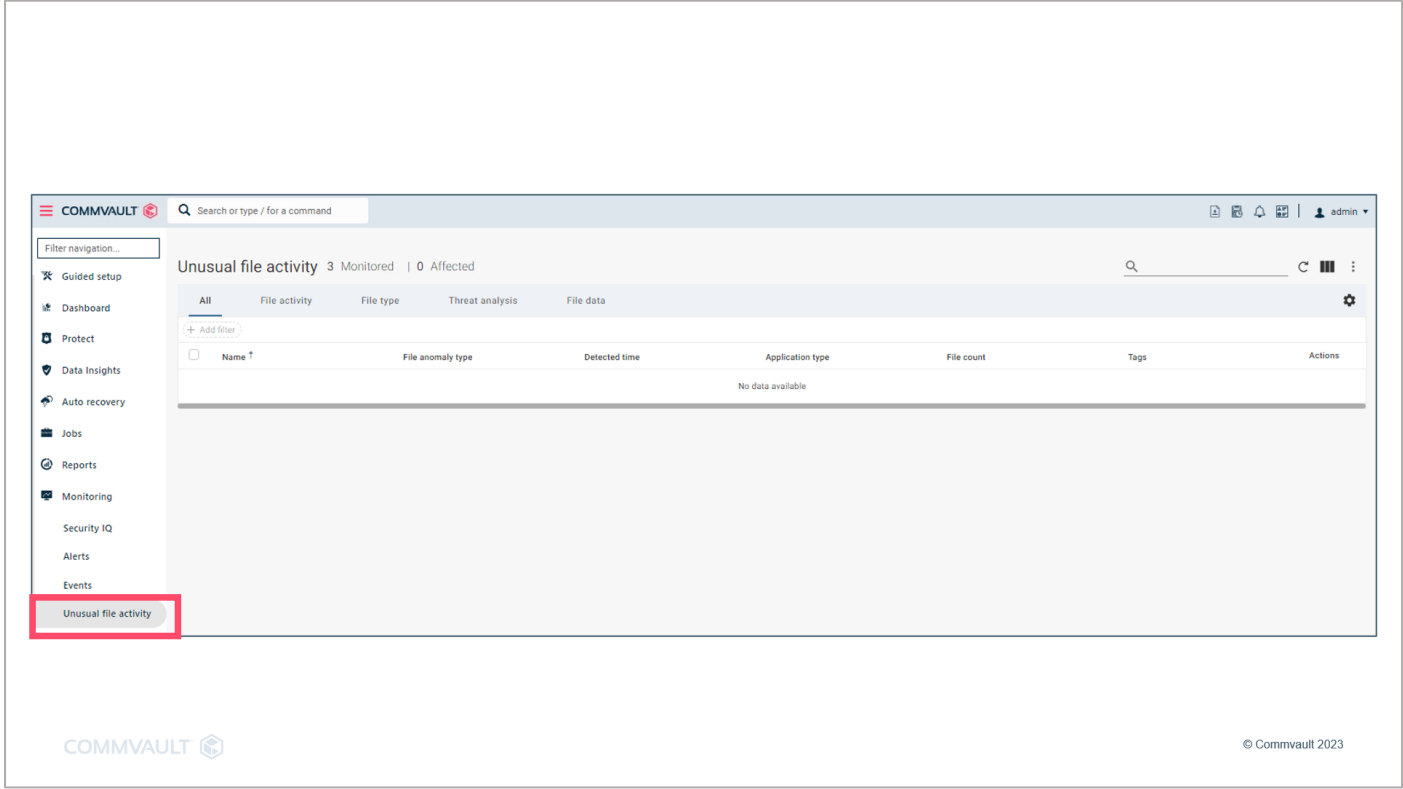
Description: Job Completed Successfully

All Commvault software related events are reported in the *Events* view. The default display is 500 events. The event log can maintain up to 10,000 events or 7 days of events. Events can be filtered by severity. *Job ID* and *Event ID* provide detailed information about the event or the job that triggered it.

1. Displays events that occurred in the CommCell environment
2. Filter events using the dropdown list
3. Search to locate events by content
4. Click a Job ID or Event ID number for detailed information

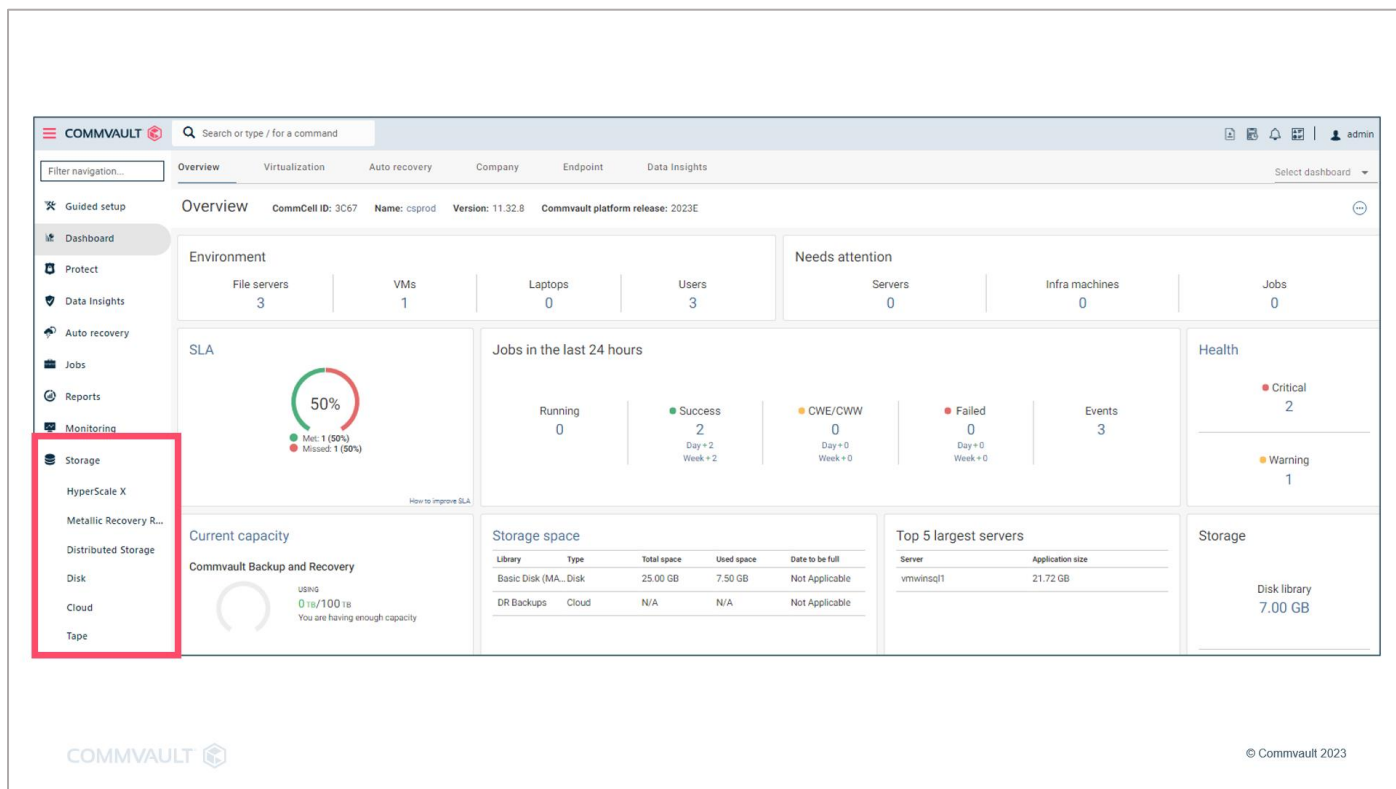
View detailed information related to the job or the event.

Unusual File Activity



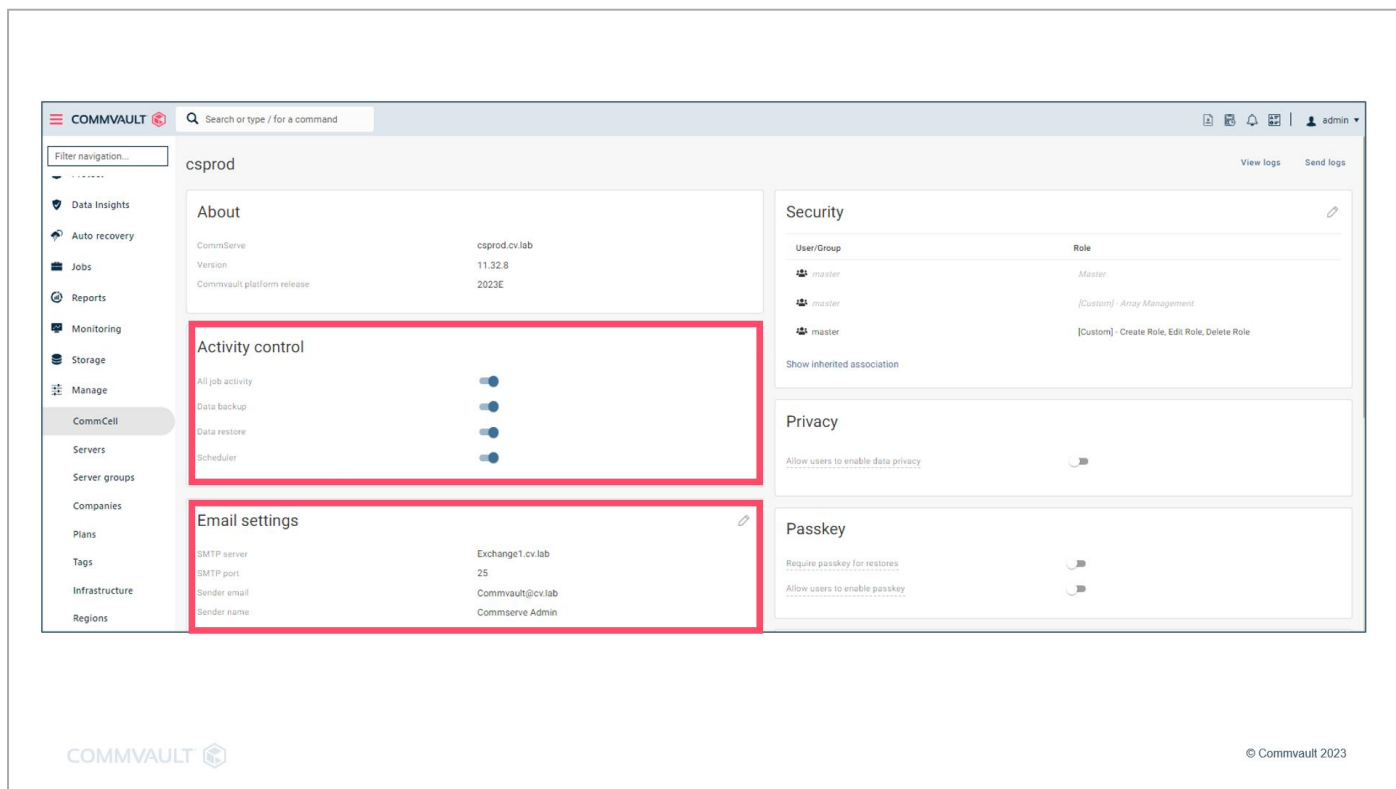
The *Unusual file activity* function displays information about anomalous activity on the file systems of active client computers and in backup jobs. This provides a single location for identifying and acting on potential threats with quick and safe recovery options. View file path information for the anomalies and track anomaly trends. Recover the most recent “good” version of an anomalous file or recover the entire client as a virtual machine from before the anomalous behavior.

Storage



The *Storage* section is for managing all types of CommCell storage.

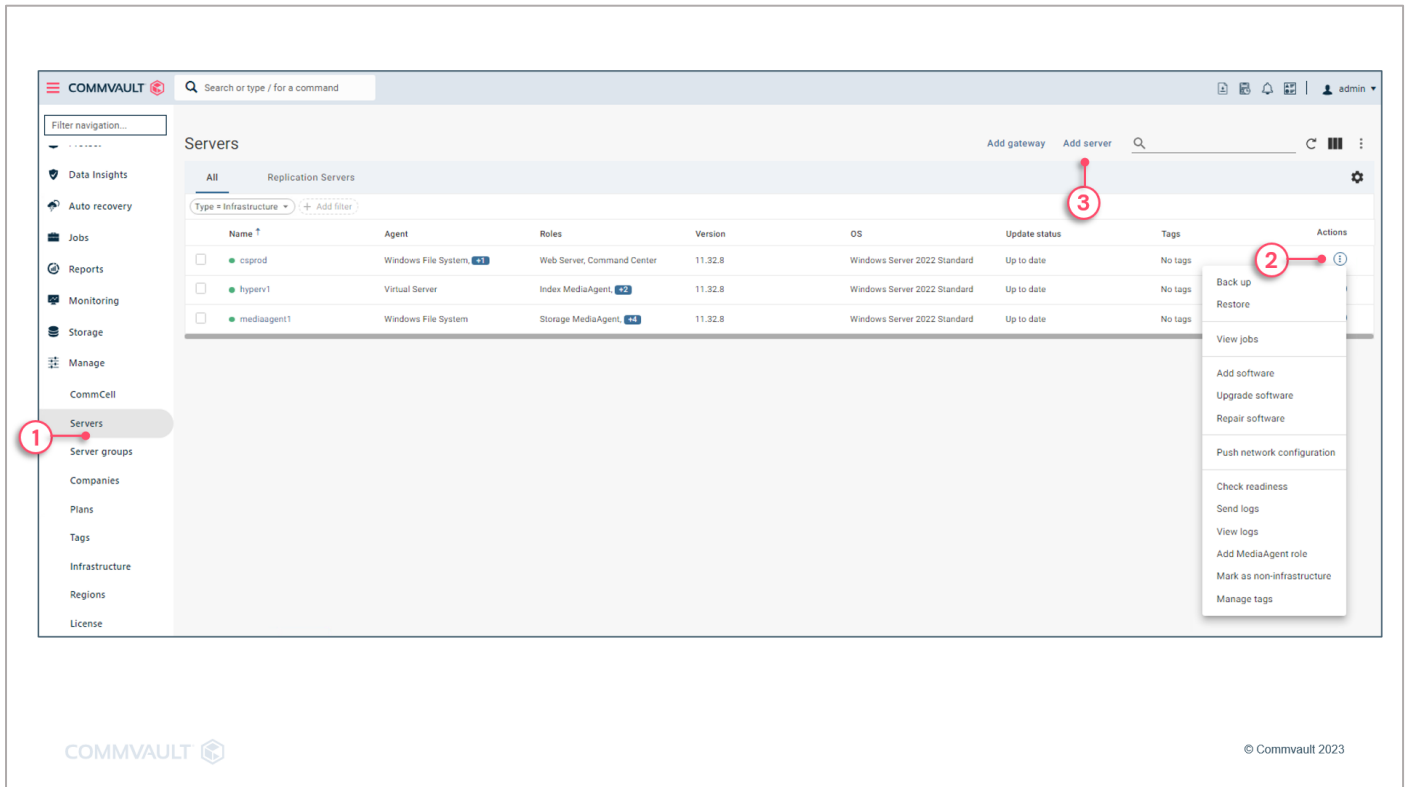
- HyperScale X
- Metallic Recovery Reserve
- Distributed Storage
- Disk
- Cloud
- Tape



The *CommCell* view provides tools to configure or control CommCell settings such as:

- **Activity Control** – enable and disable one or more activities for a CommCell environment
- **Email Settings** – configure the connection information of the organization email server that is used to relay emailed reports or alerts on behalf of Commvault software

Servers



The Servers view displays a global view of all servers on which Commvault software is installed.

1. Select **Servers**; servers are displayed in the main window
2. Click the icon for additional tasks
3. Click **Add server** to install software on new servers

Server Groups

The screenshot shows the Commvault web interface. On the left, the navigation sidebar has 'Server groups' selected, indicated by a red circle with the number 1. The main content area is titled 'Server groups' and contains a table with columns: Name, Association, Tags, and Actions. The table lists several server groups, including 'Automatic TPM Clients', 'DB servers', 'HyperV_L1', 'Infrastructure', 'Laptop Clients', 'Media Agents', 'Proxy Clients', and 'Win FS Servers'. A context menu is open for the 'Automatic TPM Clients' group, with the 'Actions' icon highlighted by a red circle with the number 2. The context menu includes options such as 'Edit associations', 'View jobs', 'Add software', 'Upgrade software', 'Repair software', 'Uninstall software', 'Clone', 'Push network configuration', 'Restart services', 'Send logs', 'Delete', and 'Manage tags'.

Name	Association	Tags	Actions
Automatic TPM Clients	Manual	No tags	
DB servers	Manual	No tags	
HyperV_L1	Manual	No tags	
Infrastructure	Automatic	No tags	
Laptop Clients	Automatic	No tags	
Media Agents	Automatic	No tags	
Proxy Clients	Automatic	No tags	
Win FS Servers	Automatic	No tags	

The *Server groups* view displays all existing client computer groups. Click the client computer group name to display configuration options. The *Actions* icon provides access to options such as deleting the group.

1. Select *Server groups* to display all server groups in the CommCell environment
2. Click the **Action icon** for additional tasks

Plans

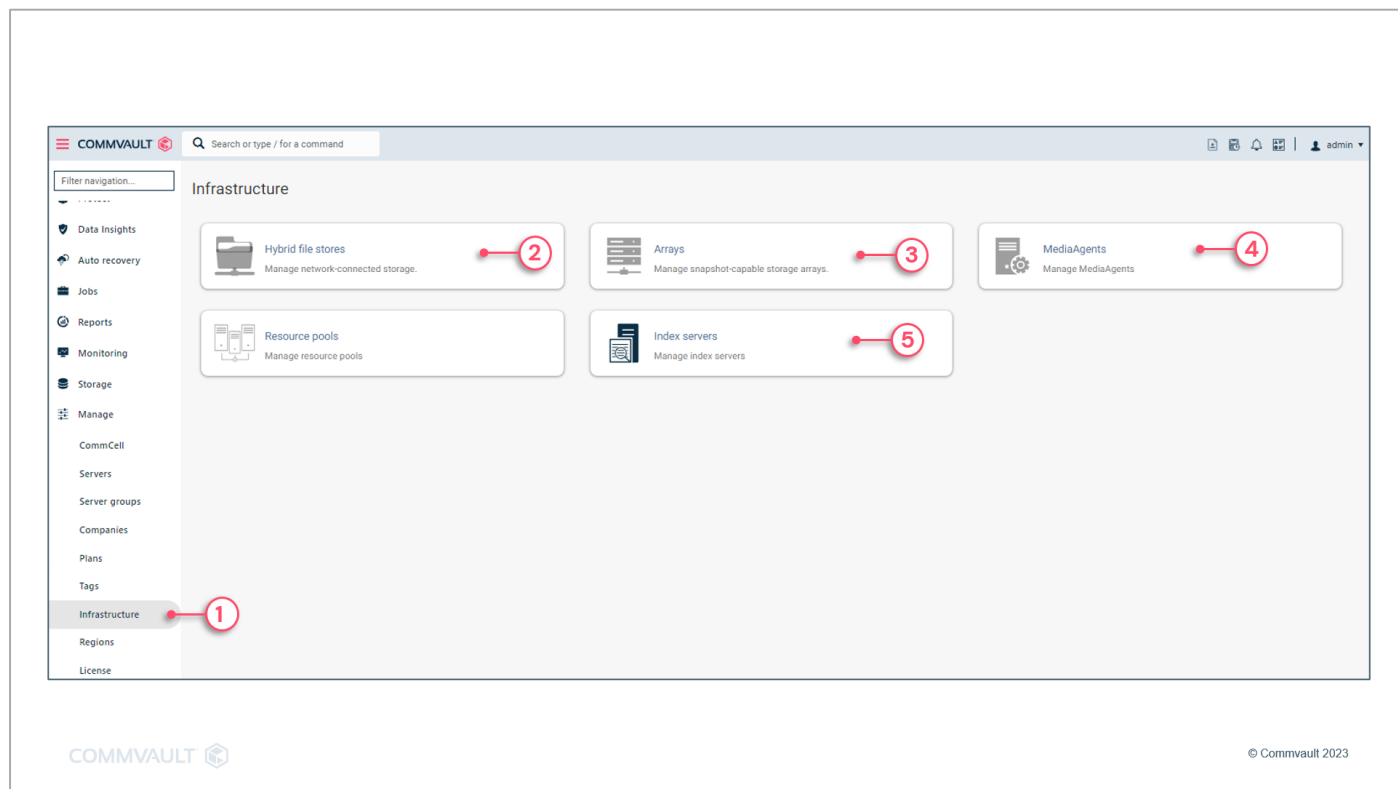
The screenshot shows the Commvault web interface. On the left is a navigation sidebar with a search bar and a list of categories: Data Insights, Auto recovery, Jobs, Reports, Monitoring, Storage, and Manage. The 'Plans' option under 'Manage' is highlighted with a red circle and the number 1. The main window displays the 'Plans' view. At the top right of the main window, there are links for 'Plan rules' and 'Create plan', with the 'Create plan' link highlighted by a red circle and the number 3. Below these links is a table of backup plans. The table has columns for Plan name, Plan type, Associated entities, RPO, Number of copies, Status, Tags, and Actions. Three plans are listed: Desktop Laptop Plan, Production Server Plan, and Server Backup Plan. The Actions column for each plan contains a red circle with a white ellipsis icon, which is highlighted by a red circle and the number 2. The Commvault logo and copyright notice '© Commvault 2023' are at the bottom.

Plan name ↑	Plan type	Associated entities	RPO	Number of copies	Status	Tags	Actions
Desktop Laptop Plan	Server	0	2 days	2	Enabled	No tags	⋮
Production Server Plan	Server	7	1 day	2	Enabled	No tags	⋮
Server Backup Plan	Server	0	1 day	2	Enabled	No tags	⋮

The *Plans* view displays all existing backup plans created in the CommCell environment. Use this view to add, delete, or modify a plan.

1. Select *Plans*, Plans are displayed in the main window
2. Click the icon for additional tasks
3. Click **Create plan** to create a new backup plan

Infrastructure



The *Infrastructure* view displays additional CommCell components.

1. Select **Infrastructure** to display the infrastructure components of the CommCell environment
2. Click the **Hybrid file stores** tile to configure hybrid file stores (network stores)
3. Click the **Array** tile to configure storage arrays that are protected using the IntelliSnap feature
4. Click the **MediaAgents** tile to configure MediaAgent settings
5. Click the **Index servers** tile to configure index servers for File Storage Optimization, Data Governance, and eDiscovery & Compliance

Regions

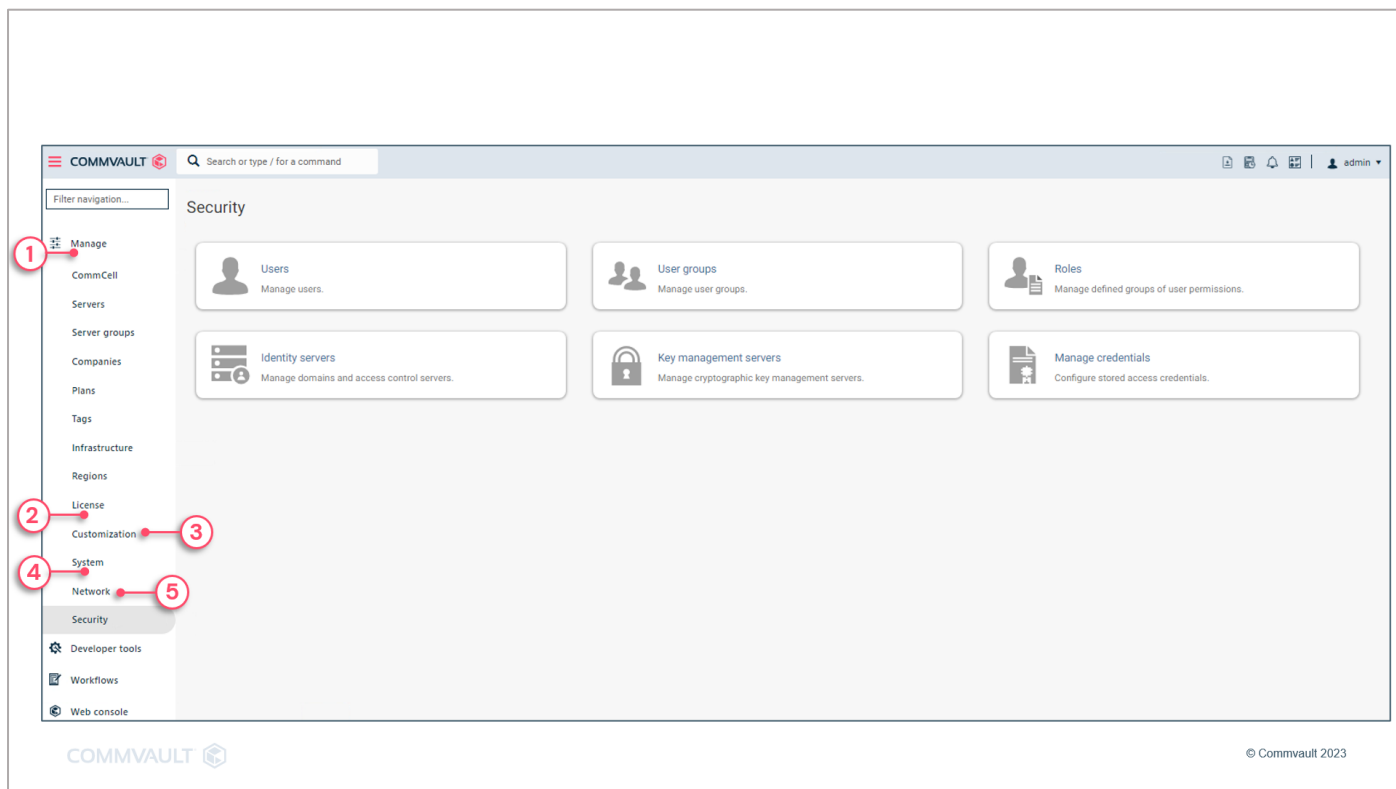
The screenshot displays the Commvault web interface for managing regions. The left sidebar contains navigation options: Data Insights, Auto recovery, Jobs, Reports, Monitoring, Storage, Manage, CommCell, Servers, Server groups, Companies, Plans, Tags, Infrastructure, **Regions** (highlighted with a red circle 1), and License. The main content area shows the 'Regions' section with a table of existing regions. A modal form for adding a new region is open, showing fields for Region name (3), Region type (4), and Location (5). The table lists regions such as 'af-johannesburg-1', 'af-south-1', 'Africa', 'ap-chuncheon-1', 'ap-east-1', 'ap-hyderabad-1', 'ap-melbourne-1', 'ap-mumbai-1', 'ap-northeast-1', 'ap-northeast-2', 'ap-northeast-3', 'ap-osaka-1', and 'ap-seoul-1'. The modal form includes a 'CANCEL' button and a 'SAVE' button.

Region	Region type	Location	Associated servers	Associated region based plans	Actions
af-johannesburg-1	OCI	Johannesburg-Gauteng-South Africa	0	0	
af-south-1	AWS	Default	0	0	
Africa	Default	Algeria, Angola, Benin, Botswana, CD	0	0	
ap-chuncheon-1	OCI	Chuncheon-Gangwon-do South Korea	0	0	
ap-east-1	AWS	Default	0	0	
ap-hyderabad-1	OCI	Hyderabad-Telangana-India	0	0	
ap-melbourne-1	OCI	Melbourne-Victoria-Australia	0	0	
ap-mumbai-1	OCI	Mumbai-Maharashtra-India	0	0	
ap-northeast-1	AWS				
ap-northeast-2	AWS				
ap-northeast-3	AWS				
ap-osaka-1	OCI				
ap-seoul-1	OCI				

The *Regions* section allows you to define regions used in an elastic plan. An elastic plan is used to protect geographically scattered clients (e.g., roaming users' laptops) by sending the data to the closest MediaAgent storage target. This simplifies plan management by consolidating plans geographically.

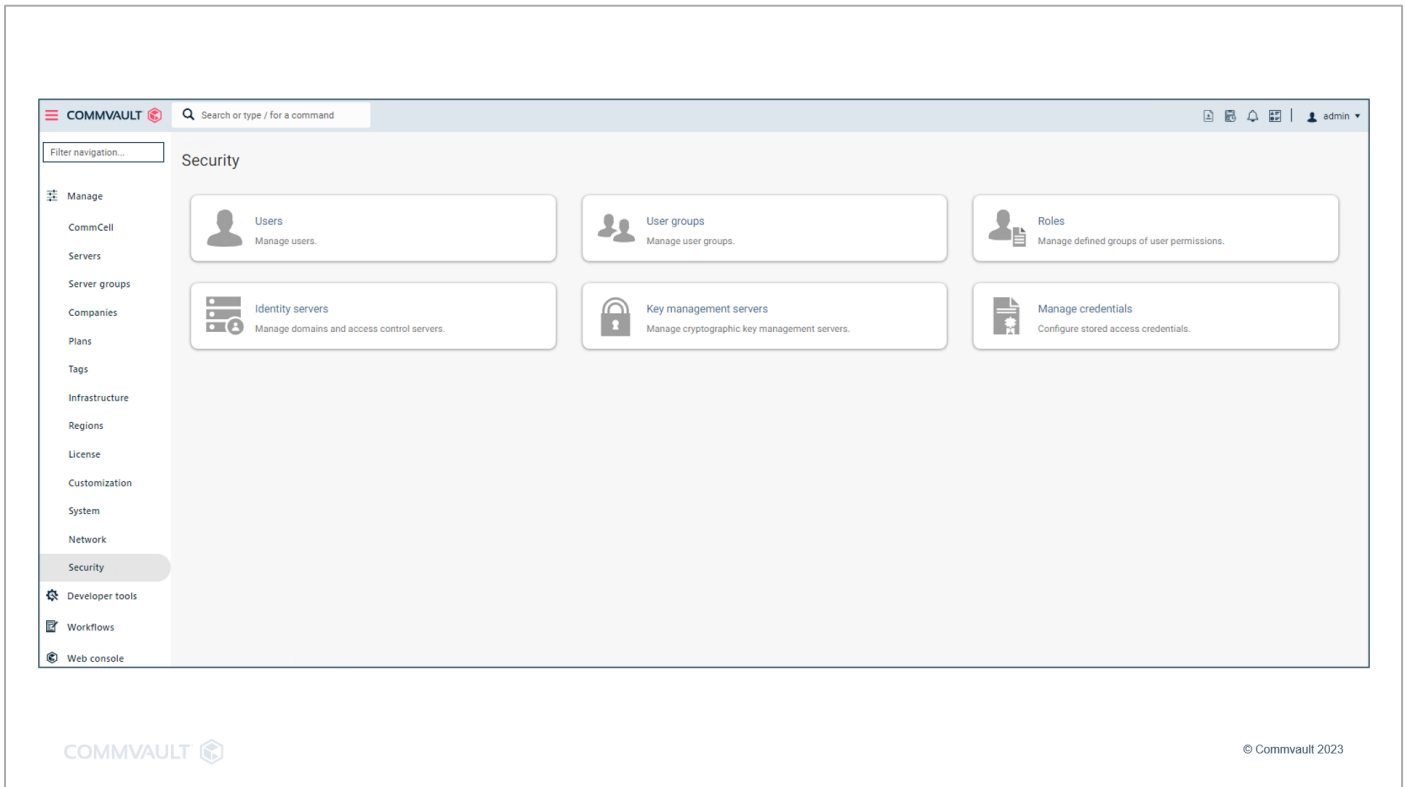
1. Select **Regions** to display all defined regions used by elastic plans
2. Click **Add region** to define a region
3. Enter a region name
4. Enter the location
5. Click **Save**

Additional Administrative Views



1. Select **Manage** to access additional administrative views
2. Select **License** to access licensing information and apply a license file
3. Select **Customization** to configure the Commvault Command Center layout and colors
4. Select **System** to download software and configure system settings such as global exceptions, system owner permissions, and DR backups
5. Select **Network** to configure network components such as network topologies and data interface pairs

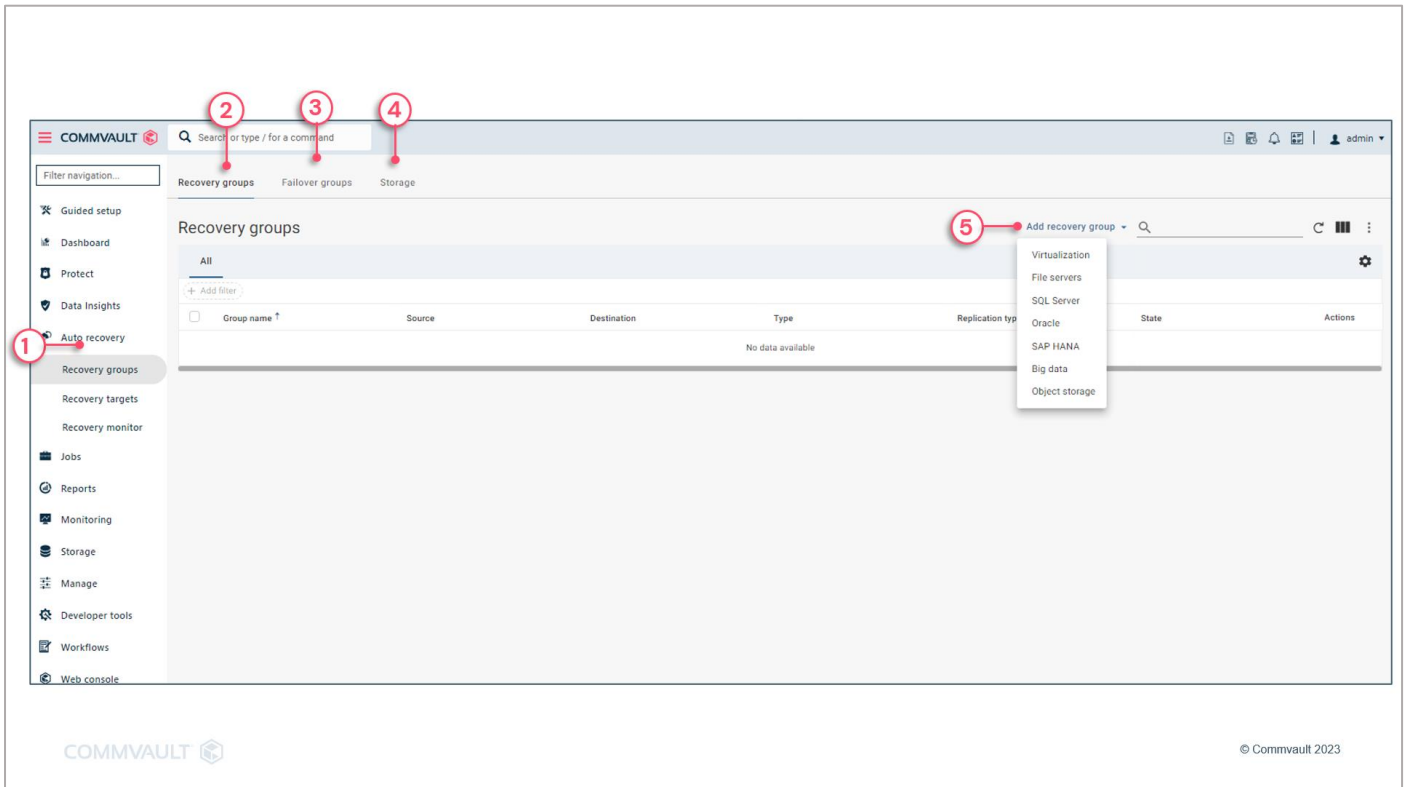
Security



The *Security* section allows a user to create and manage users, user groups, and domains that are used in security associations. The security associations define user access on CommCell entities. The following security components can be configured:

- **Users** – manage your Commvault users
- **User groups** – manage your Commvault user groups
- **Roles** – manage permissions that are assigned to users through a security association
- **Identity servers** – manage domains that are integrated with Commvault software to leverage domain user accounts and user groups rather than CommCell users and user groups, and manage the integration of third-party identity providers
- **Key management servers** – manage third-party key management servers managing hardware and/or software encryption keys
- **Manage Credentials** – locate all credentials required to access resources such as NAS or cloud storage

Replication and Auto Recovery



Auto Recovery

Synchronize virtual machines and filesystems from a source system to a target infrastructure, including remote sites.

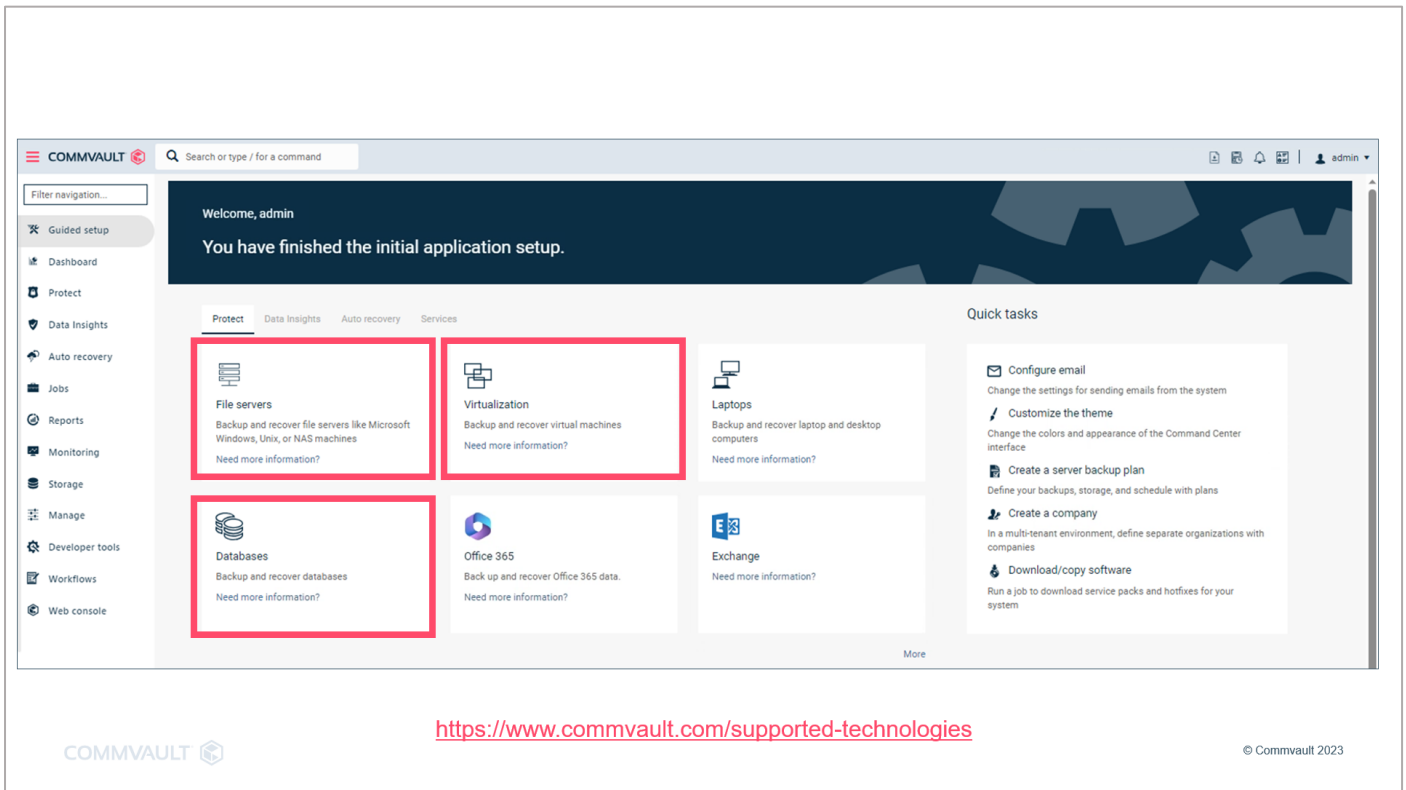
- **Recovery Groups** – Configure Commvault Live Sync replication for groups of virtual machines. This efficient method prevents the Commvault administrator from having to configure replication pairs, one VM at a time. This view displays configured groups and lets you add VM groups.
- **Failover Groups** – Commvault software implements disaster recovery orchestration for virtual machines. If a VM or a group of VMs suffer from any type of disaster, Commvault software provides mechanisms to failover the virtual machines to a secondary site. This feature requires the implementation of Commvault Live Sync. The Failover Groups view displays the configured recovery groups and lets you add groups.
- **Storage** – The *Storage section* allows configuration of recovery point stores on the destination site when replicating virtual machines using Live Sync I/O. For each replication pair, application-consistent recovery points and updates are recorded in the store, allowing VMs to be recovered as needed.

Replication

1. Select *Auto Recovery* to configure replication and failover for different systems
2. Click the *Recovery groups* tab to view replication groups and configuration options
3. Click the *Failover groups* tab to configure virtual machine failover groups using DR orchestration
4. Click the *Storage* tab to view configured replication targets (stores) for Live Sync I/O replication pairs
5. Click **Add recovery group** to configure a new recovery group

Features

Features Overview



Features make it easy to manage day-to-day tasks such as monitoring backup infrastructure, performing the backup and recovery of computers, servers and virtual machines, and protecting cloud applications. Features are published to end users or system administrators granularly using Commvault software role-based security.

File Servers Feature

The File Servers feature provides tools to manage protection for UNIX and Windows servers. Once the Commvault Command Center core setup is complete, the file servers feature can be configured. Clients and subclients are configured directly from the Commvault Command Center.

Commvault software provides the following features so that you can efficiently manage your data protection, using the default settings or customizing the settings for your needs.

- Job monitoring
- Alerts
- Events
- Resource reports

Virtualization Feature

The Virtualization feature provides a simplified management interface for administrators to conduct day-to-day operations to configure, protect and recover virtual machines using the Virtual Server Agent (VSA).

Use the Commvault Command Center's Virtualization dashboard to check the status of hypervisors, such as VMware or Hyper-V. Check the number of hypervisors in your environment and the number of VMs protected. The virtualization dashboard also includes backup job summaries.

Hypervisors and subclients can be configured directly from the Commvault Command Center. Use the default plan or configure your own settings to dictate when VMs are backed up and where the data is stored. You can add backup plans as needed.

Databases Feature

The databases feature can be configured after the Commvault Command Center Core Setup is complete. Configure database instances and subclients directly from the Commvault Command Center. A backup plan dictates when databases are backed up and where the data is stored. Create a backup plan during configuration or use the default plan. Additional backup plans can later be defined.

Go to <https://www.commvault.com/supported-technologies> to view the current list of supported workloads.

Commvault Command Center Customization

Define Columns to Display in a View

The screenshot displays the Commvault Command Center interface. The left sidebar contains navigation options: Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs (selected), Reports, Monitoring, Storage, Manage, Developer tools, Workflows, and Web console. The main area shows the 'Job history' view with a table of jobs. The table has columns: Job id, Operation, Server, Agent type, Subclient, Size, End, Elapsed, Status, Error description, and Actions. A custom column configuration menu is open, showing options to select or deselect columns. The 'Agent type' column is selected, and the 'Backup copy job ID' column is also selected. The 'Backup copy status' and 'Backup type' columns are deselected. The 'RESET' and 'SAVE' buttons are visible at the bottom of the menu.

Job id	Operation	Server	Agent type	Subclient	Size	End	Elapsed	Status	Error description	Actions
13	Auxiliary Copy				0.00 B	Apr 4, 2023, 2:03:48 ...	5 days 3 hours 37 m...	Killed	Killed	
20	Auxiliary Copy				227.13 MB	Apr 4, 2023, 5:06:58 ...	2 hours 27 minutes ...	Killed	Killed	
11	Data Aging				0.00 B	Mar 29, 2023, 12:08:...	0 seconds	Failed to Start	Failed to Start	
18	Data Aging				0.00 B	Apr 4, 2023, 12:00:0...	0 seconds	Failed to Start	Failed to Start	
22	Data Aging				0.00 B	Apr 5, 2023, 12:00:0...	0 seconds	Failed to Start	Failed to Start	
24	Data Aging				0.00 B	Apr 6, 2023, 12:00:0...	0 seconds	Failed to Start	Failed to Start	
26	Data Aging				0.00 B	Apr 26, 2023, 7:02:5...	0 seconds	Failed to Start	Failed to Start	
28	Data Aging				0.00 B	Apr 28, 2023, 12:00:...	0 seconds	Failed to Start	Failed to Start	
29	Data Aging				0.00 B	Jul 26, 2023, 12:00:0...	0 seconds	Failed to Start	Failed to run Data A...	
35	Install Software				0.00 B	Sep 7, 2023, 6:53:23...	6 seconds	Failed	Failed to install File ...	

Customize Views

Several Commvault Command Center sections offer the option to create customized views for different user roles. For instance, a view can be created for a UNIX system administrator to display only UNIX and Linux servers. The custom view can also be defined as the default view for a section. To create the custom view, open the section from which it will be created, click to display options, and choose Create view.

Customize Navigation

The Commvault Command Center navigation is customized based on the type of users logged in.

- **Navigation preferences** – Defines the sidebar entities for each type of users
- **Default landing page** – The main dashboard is the default; this landing page can be changed to another view

Customize Theme

The Commvault Command Center can be branded using your company's colors and logo. This gives a personalized experience to your users. This option is also frequently used by Managed Services Provider (MSP) to provide their personalized console to their Backup-as-a-Service clients. Change background colors, add a background image, and add a company logo.

To customize the Commvault Command Center colors, select **Manage > Customization** from the sidebar and click **Theme**. In any view, the columns that are displayed can be modified, added, or removed. Changes are saved to the user's profile.

1. Click the **Actions** icon and select **Column**
2. Select or deselect columns to display

Apply Filters to a View

The top screenshot shows the 'Job history' view with a 'Filters' dialog box open. The dialog has a 'Filter by' dropdown set to 'Operation'. Below it, a search bar contains 'Auxiliary ...'. A list of operations is shown, with 'Auxiliary Copy' selected. The bottom screenshot shows the 'Job history' table with the 'Operation' column header dropdown menu open. The filter 'Operation = Auxiliary Copy' is applied, and the table shows only jobs with that operation. The table has columns: Job Id, Operation, Server, Agent type, Subclient, Size, End, Elapsed, Status, Error description, and Actions. The status column shows 'Killed' for two jobs and 'Completed' for one.

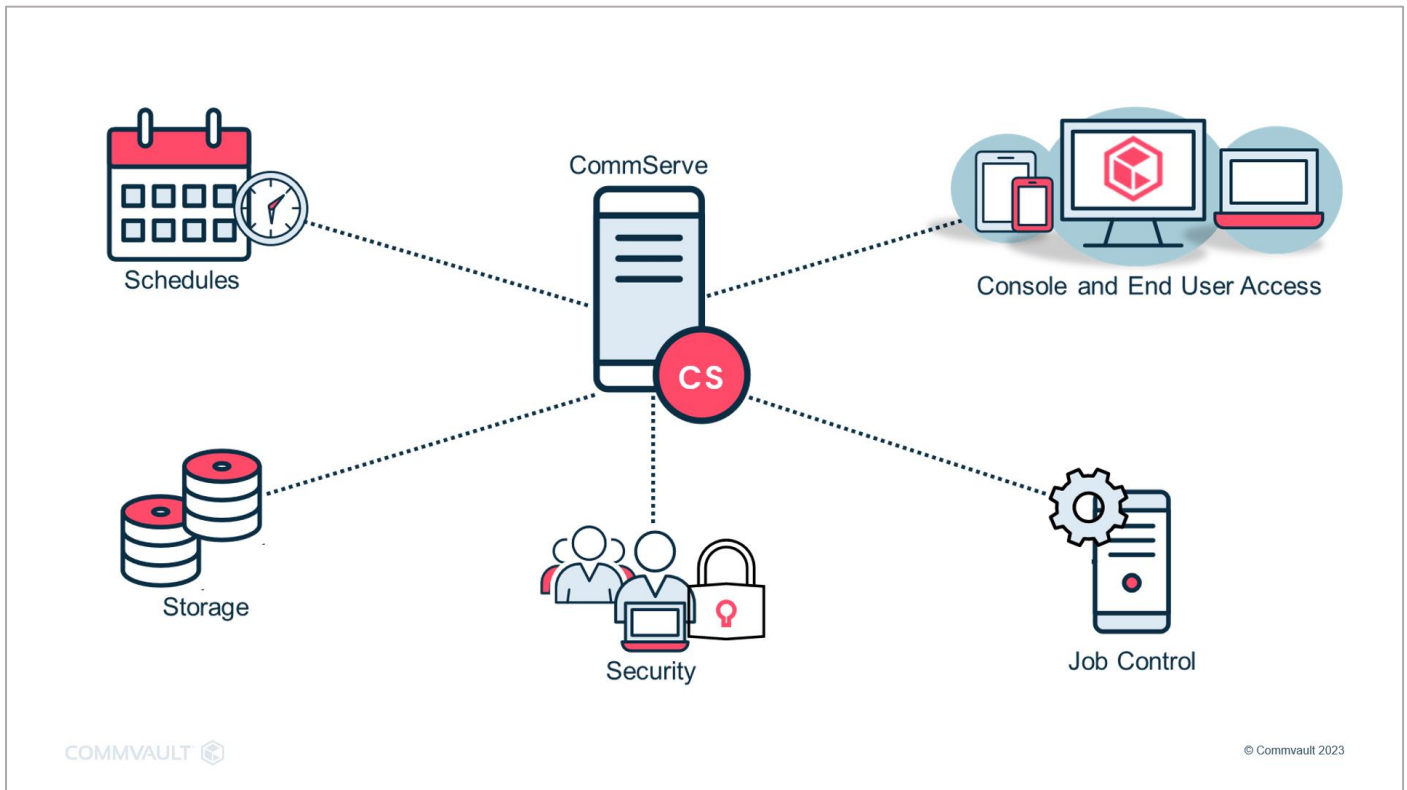
Job Id	Operation	Server	Agent type	Subclient	Size	End	Elapsed	Status	Error description	Actions
19	Auxiliary Copy				0.00 B	Apr 4, 2023, 2:03:48 ...	5 days 3 hours 37 m...	Killed	Killed by admin. Rea...	⋮
20	Auxiliary Copy				227.13 MB	Apr 4, 2023, 5:06:58 ...	2 hours 27 minutes ...	Killed	Killed by admin. Rea...	⋮
10	Auxiliary Copy				2.36 GB	Mar 29, 2023, 11:21:...	1 minutes 51 seconds	Completed		⋮

Filters can be applied to most views. The option to apply filters must be toggled ON before selecting the filters from the column header dropdown list.

1. Click the **Actions** icon for the desired column and click **Filter**
2. Click the **Condition** field and select the entries to display
3. Click **OK** to apply it to the view
4. Click **Clear all filters** to return to the default view

CommCell Core Components

CommServe Server



The CommServe server is the central management system within a CommCell environment. All activity is coordinated and managed by the CommServe server. The CommServe server must be scaled appropriately for the size of the environment. For current scalability guidelines, refer to Commvault Documentation.

CommServe Server Options

- **The CommServe server can be clustered** – This is recommended for larger environments where high availability is critical
- **The CommServe server can be virtualized** – This is suitable for small to mid-size environments

CommServe Server Performance Requirements

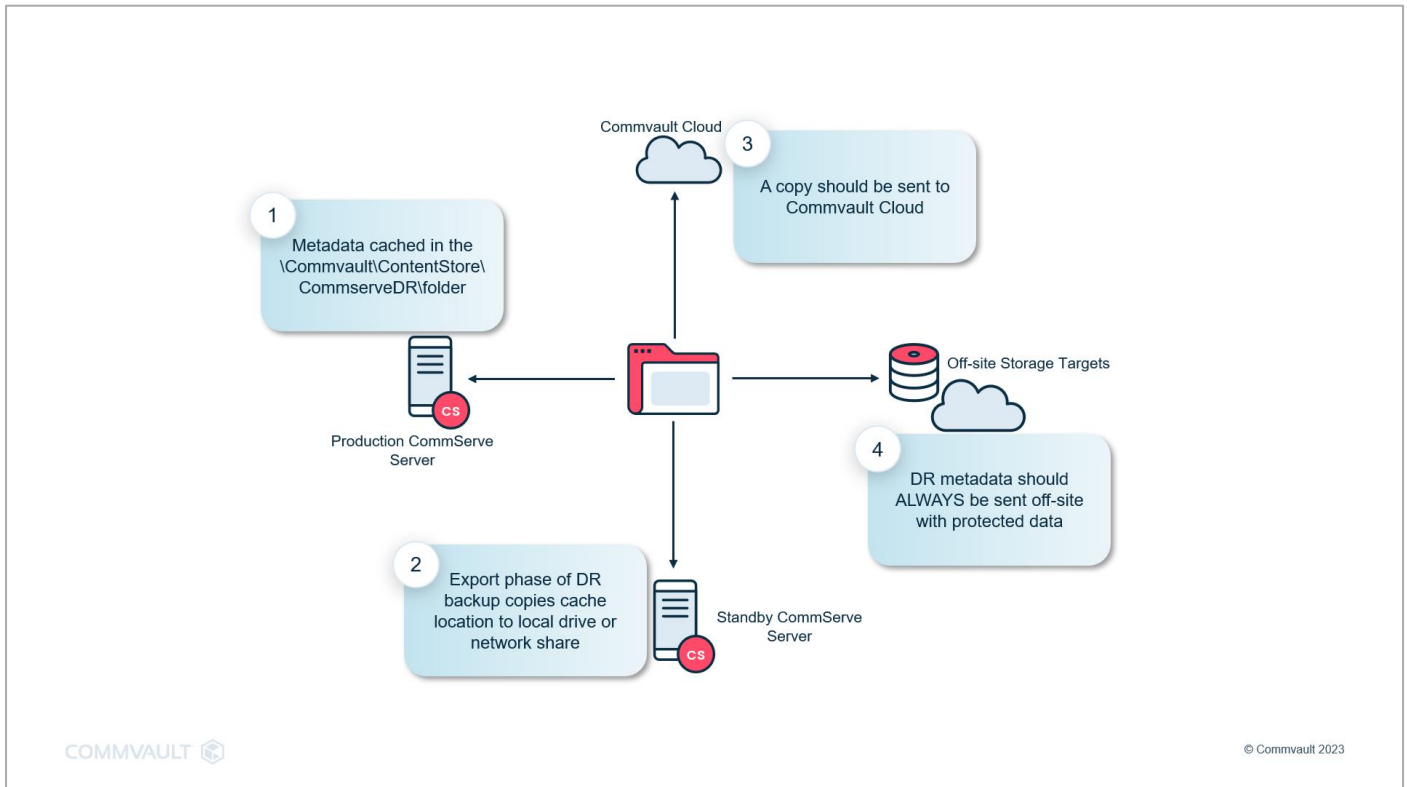
CommServe Server performance is essential for your data protection environment. Communication and job checkpoints are constantly occurring between CommCell components and the CommServe server. The CommServe server also serves other functions such as reporting, and the user experience may be impacted during peak periods of data protection operations.

CommServe Server Communication Services

During data protection jobs, the CommServe Job Manager (JobMgr) process initiates job operations. The CVD process is on all CommCell components and provides communication with all resources. As each chunk of a job completes it must be registered in the CommServe database before the next chunk begins.

During data protection and auxiliary copy jobs, the CommServe server has a substantial responsibility. Consider this when planning the resources for the CommServe server, especially in larger environments where hundreds of jobs will be running in parallel.

CommServe Disaster Recovery Backups



Note: It is ABSOLUTELY CRITICAL that the CommServe database is properly protected. The default setting runs the CommServe Disaster Recovery backup job at 10:00 AM daily. This operation can be completely customized and set to run multiple times a day if required.

This process first runs a SQL dump of the CommServe instance and is stored locally on the CommServe server.

Next, the contents from the DR folder are exported to a user-defined drive or UNC path. For the last phase, when you build the first storage location in the CommCell, Commvault automatically creates the CommServe DR Storage plan and subsequently backs up the DR Metadata to the first defined storage target.

If a tape target is configured, a secondary copy set is created for the CommServe DR storage plan. Also, a schedule is created to copy the job to the second storage location. The tape media containing the DR metadata should be sent offsite regularly.

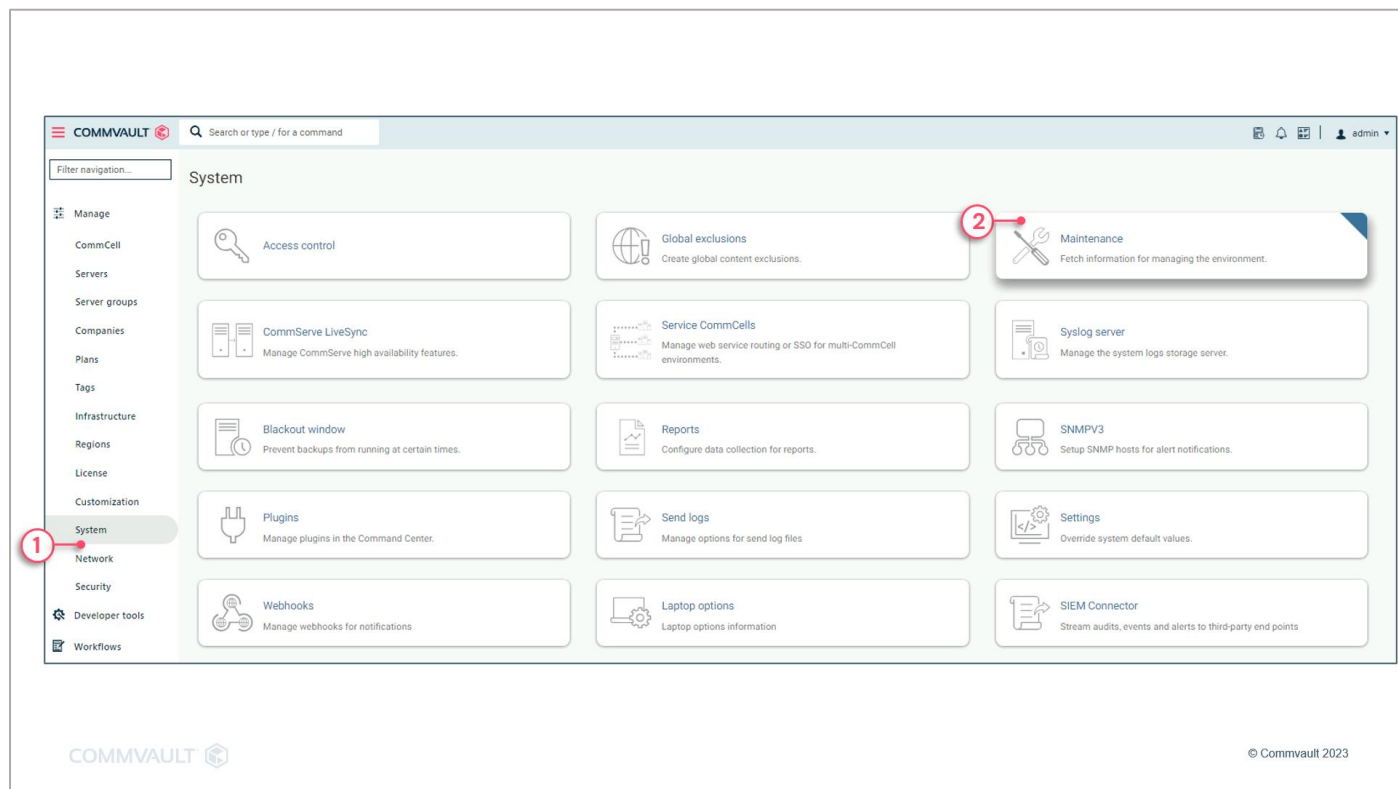
Schedules and export location are customizable in the DR Backup applet in the *Maintenance* section of the Commvault Command Center.

Export

The export process copies the content of the `\CommServeDR` folder to the user defined export location. A local drive or UNC path can be defined. The export location should NOT be on the local CommServe server. If a standby CommServe server is available, define the export location to a share on the standby server.

By default, five metadata backups are retained in the export location. It is recommended to have enough disk space to maintain one week of DR exports and adjust the number of exports to the DR backup schedule frequently.

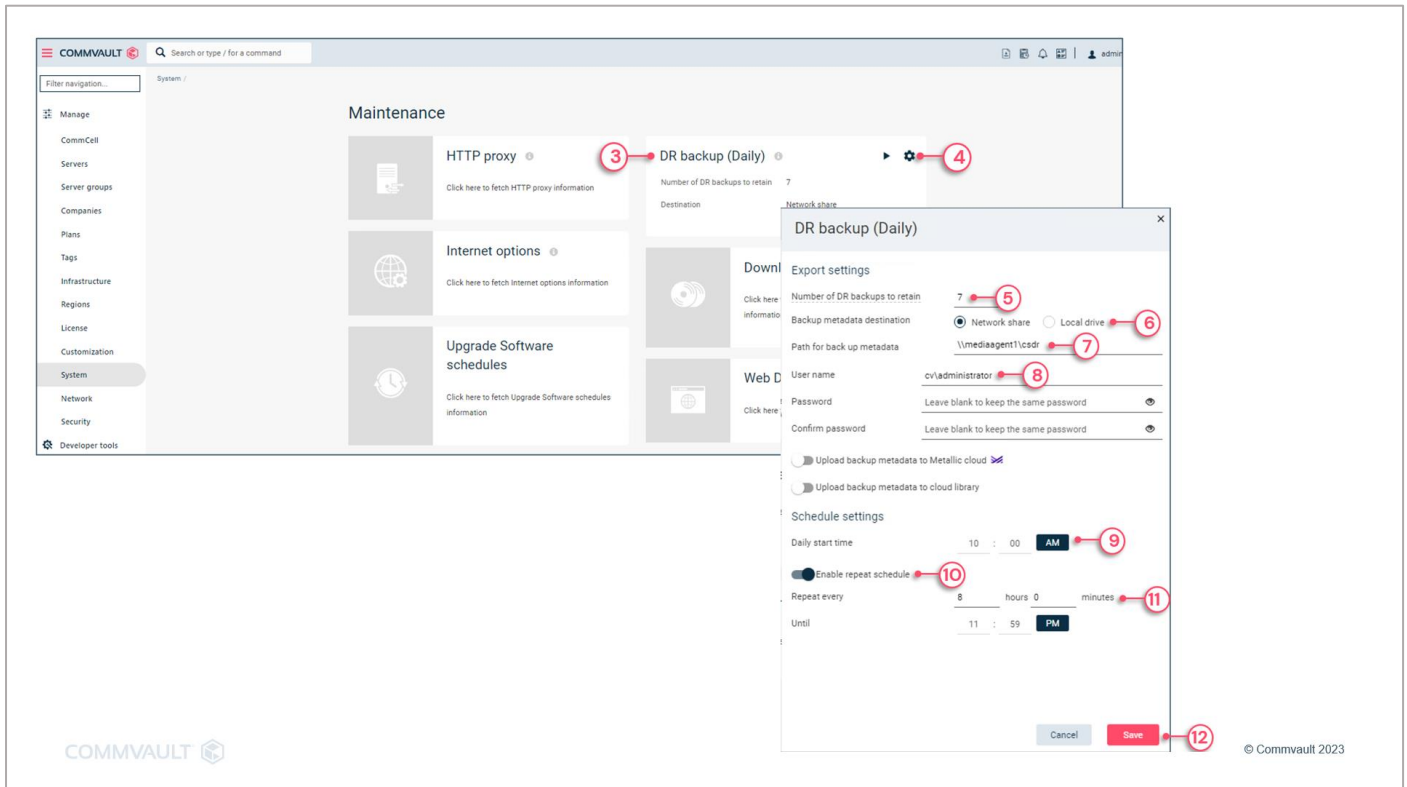
Configure and Run DR Backups



DR backups are automatically configured and scheduled when you install the software. The default settings and schedule can be edited to fit your needs, and manual DR backups can be executed on demand if needed.

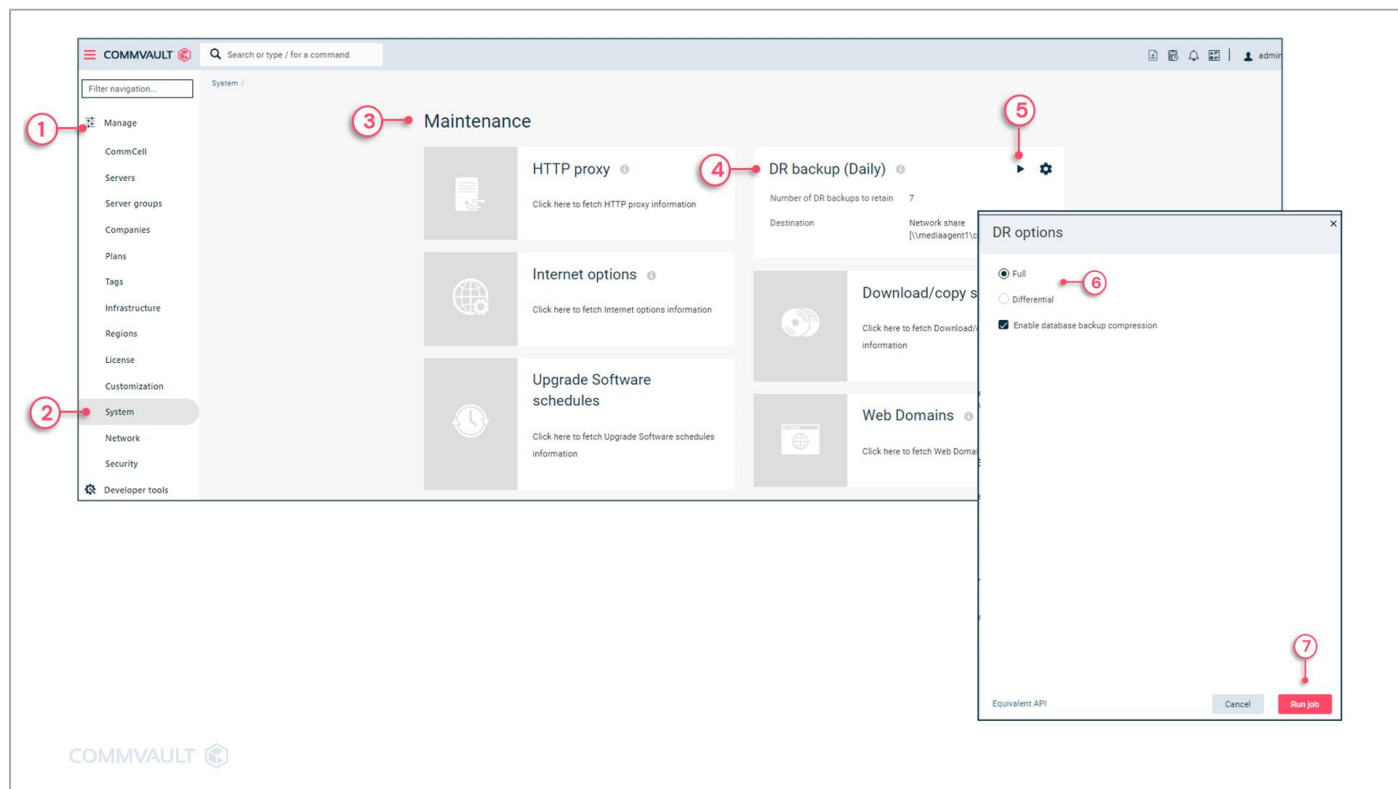
1. Select **Manage > System**
2. Click the **Maintenance** tile

Configure and Run DR Backups ...cont'd



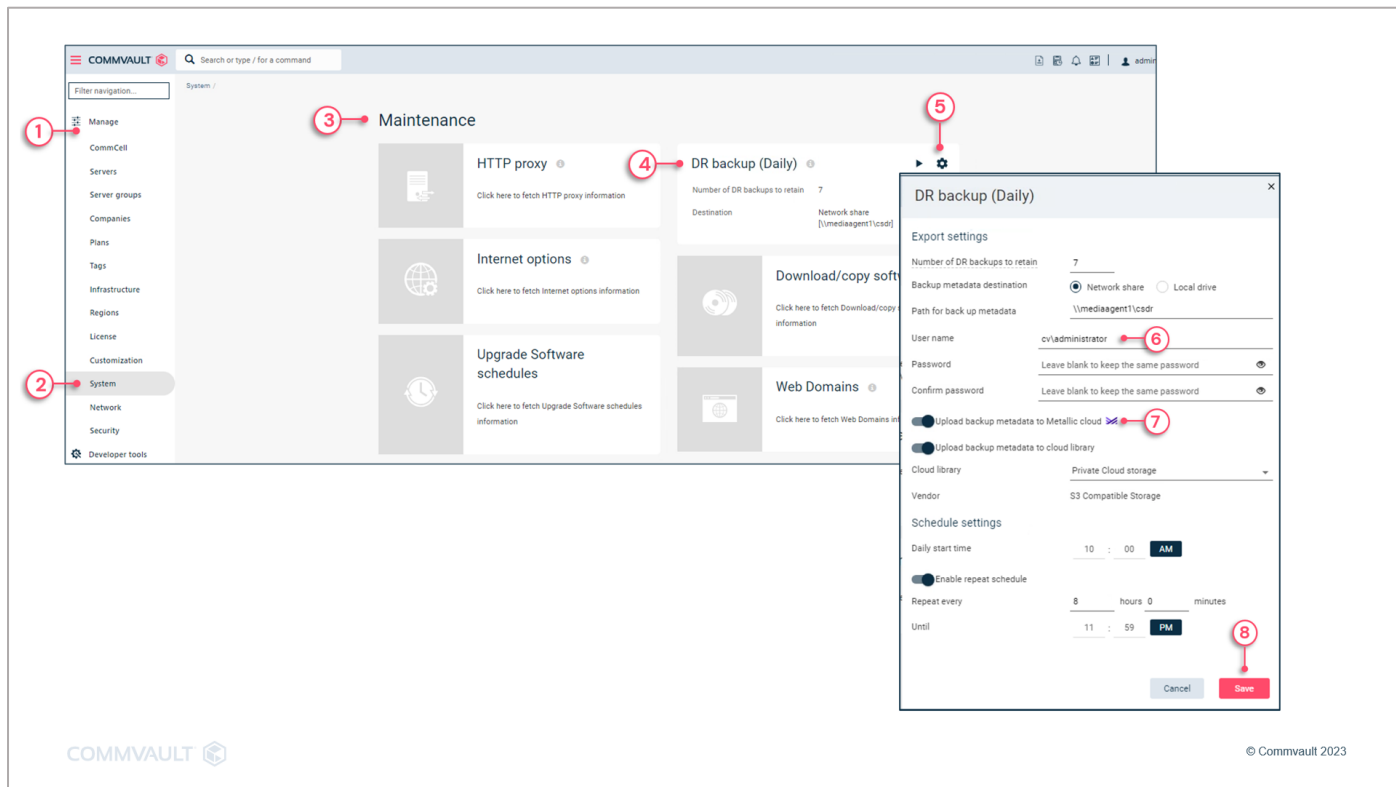
3. Locate the **DR backup (Daily)** tile to access the DR backup settings
4. Click the **Gear icon** to edit DR backup settings
5. Enter number of metadata backups to retain
6. Select the type of destination to use
7. Enter the share UNC path
8. Enter credentials with access permissions to the destination
9. Set the time of execution
10. Toggle on **Enable repeat schedule** to enable multiple DR backups to run daily
11. Set the time range for the backup schedule
12. Click **Save**

Manually Run a DR Backup



1. Select **Manage**
2. Click **System**
3. Click the **Maintenance** tile
4. Click the **DR backup (Daily)** tile to access the DR backup settings
5. Click the arrow to continue
6. Select the backup type and check the box for **Enable database backup compression** if desired
7. Click **Run job** to launch the job

DR Backups to Commvault Cloud

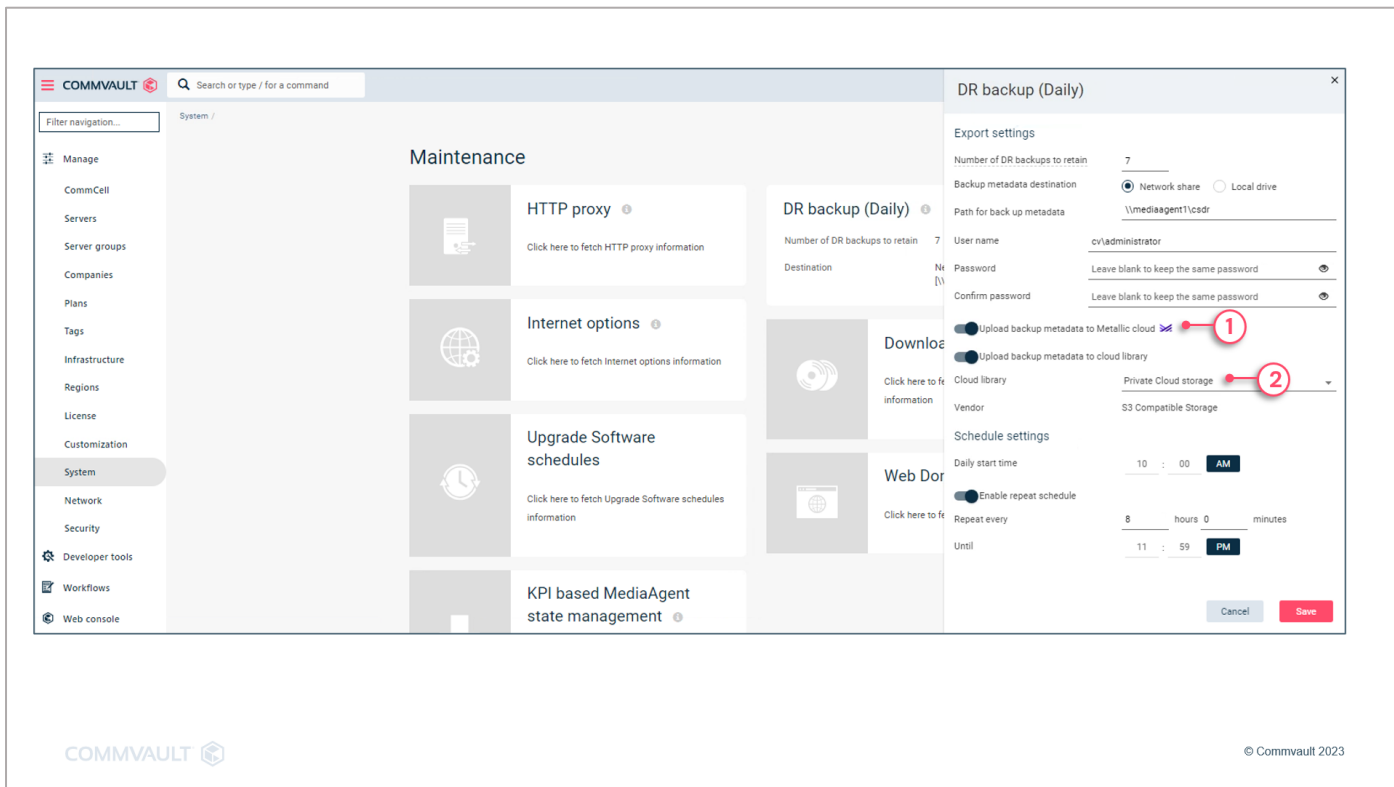


Take advantage of a free Commvault Cloud Services account to back up your Disaster Recovery data to the cloud. The service stores the last seven metadata backups. This ensures that a recent copy of the database is offsite and cannot be accessed by a rogue process such as a ransomware attack.

The free cloud service requires a Commvault Cloud Services account. Go to cloud.commvault.com for more information.

1. Select **Manage**
2. Click **System**
3. Click **Maintenance**
4. Locate the *DR backup (Daily)* section
5. Click the **Gear icon** to edit DR backup settings
6. Enter the cloud services account credentials
7. Toggle on **Upload backup metadata to Commvault cloud** to enable DR backups save to the cloud
8. Click **Save**

Save DR Backups to a Cloud Storage Target



To send a copy of the DR backup to a cloud storage target, the administrator needs to connect to the pre-established cloud storage vendor and upload the DR backup copy.

Manage > System > DR backup (Daily) section > Gear icon

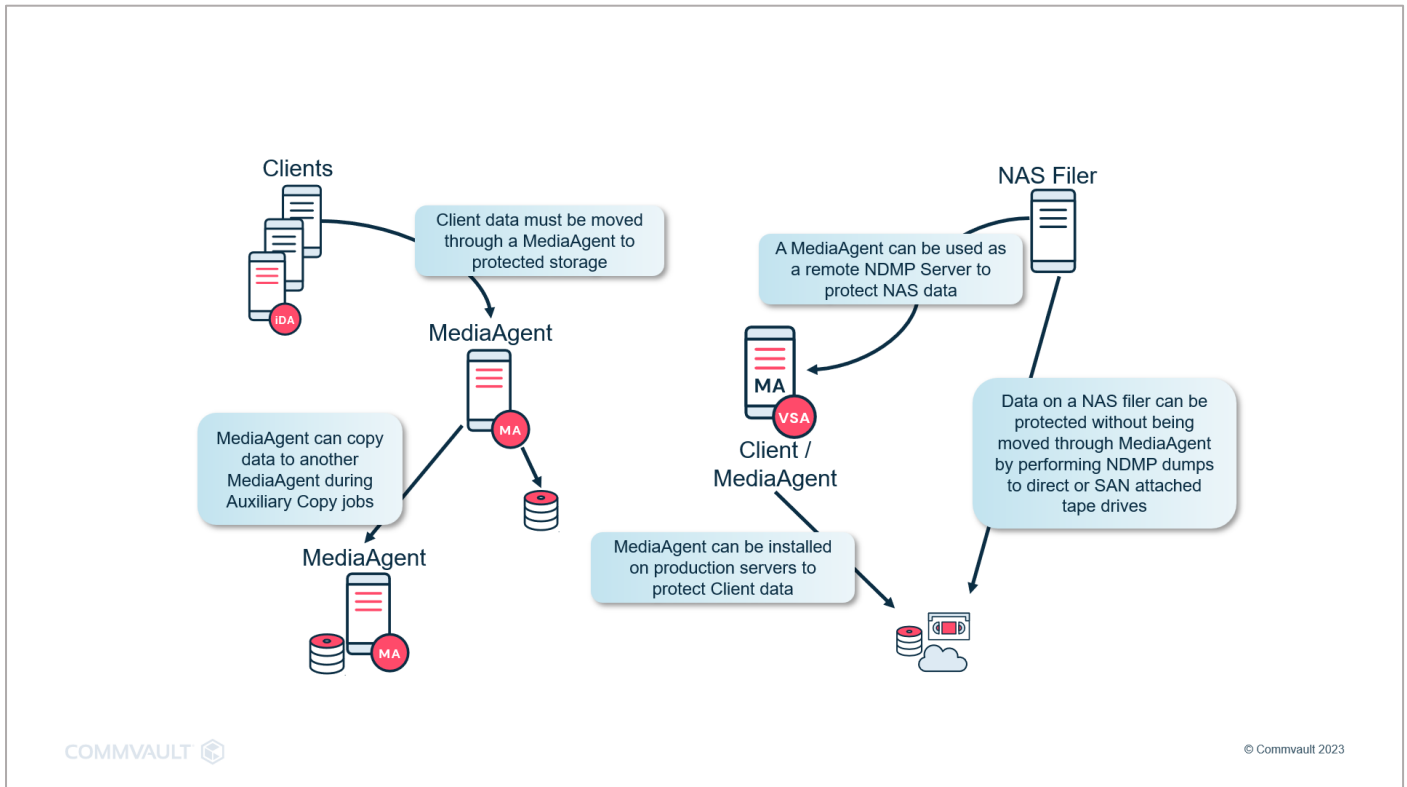
1. Toggle on **Upload backup metadata to cloud library** to send a copy to a cloud storage target
2. Select the storage target and the vendor

Backup Frequency

By default, the DR backup runs every day at 10:00 AM. The time the backup runs can be modified, and the DR backup can be scheduled to run multiple times a day.

The DR metadata is required to recover protected data. If backups are conducted at night and auxiliary copies are run during the day, consider setting up a second schedule after auxiliary copies complete.

MediaAgent



MediaAgents are the multi-function workhorses of a Commvault software environment. They facilitate the transfer of data from source to destination, hosts the deduplication database and metadata indexes, and run the analytic engines.

The MediaAgent is used during auxiliary copy jobs when data is copied from a source library to a destination library.

The MediaAgent software can be installed on most operating systems in physical, virtual, and clustered environments. All tasks are coordinated by the CommServe server.

MediaAgent and Data Movement

There is a basic rule that all data must travel through a MediaAgent to reach its destination. One exception to this rule is when conducting Network Data Management Protocol (NDMP) dumps directly to tape media.

In this case, the MediaAgent is used to execute the NDMP dump and no data travels through the MediaAgent. This rule is important to note as it affects MediaAgent placement.

Since all data moving to and from protected storage must move through a MediaAgent, resource provisioning for MediaAgent hosts (e.g., CPU, memory, bandwidth) must be adequate for both the volume and the concurrency of data movement you expect it to handle.

MediaAgent Device Control

A MediaAgent provides device control over media changers and removable media devices and writes to disk devices. This control defines the path upon which data moves to and from protected storage. In addition to normal device integrity checks, the MediaAgent can validate the integrity of data stored on the media during a recovery operation and validate the integrity of the data on the network during a data protection operation.

If the MediaAgent is located on the same host as the client agent, the exchange of data is contained within the host. This is called a SAN MediaAgent configuration, or sometimes referred to as LAN-free backups. This has advantages of keeping data off potentially slower TCP/IP networks by using local higher performance transmission devices (e.g., Fibre Channel, SCSI, etc.). On the other hand, a MediaAgent component located on a host by itself can provide dedicated resources and facilitate exchange of data over longer distances using TCP/IP (e.g., LAN, WAN, etc.).

MediaAgent Hosting Functions

The MediaAgent component also achieves additional functions other than moving data. First, the MediaAgent hosts the index directory. Every protection job allowing granular recovery must be indexed. The MediaAgent oversees job indexing and retains the indexing information in the index directory. If Commvault deduplication is enabled on a disk or a cloud library, the MediaAgent also hosts the deduplication database containing the deduplication information. Finally, if Data Analytics is in use, it requires the Analytics Engine to be installed on the MediaAgent.

Metadata Indexes

Commvault software uses a distributed indexing structure that provides enterprise level scalability and automated index management. The CommServe database only retains job-based metadata such as chunk information to keep the database relatively small. Detailed index information, such as details of protected objects, is kept on the MediaAgent. Ensure the index location is on high-speed dedicated disks.

Physical vs. Virtual MediaAgent

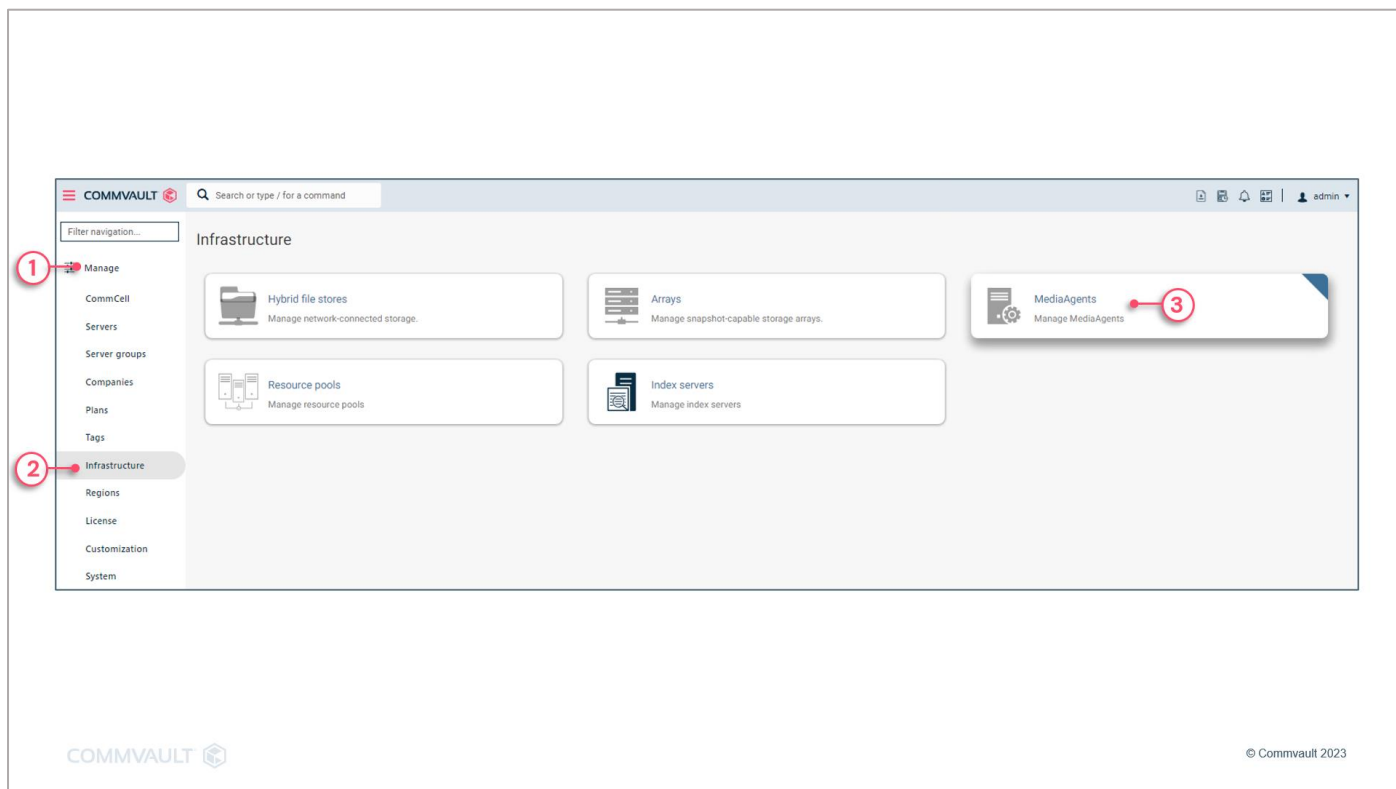
Commvault recommends using physical MediaAgents to protect physical and virtual data. The advantages for using a physical MediaAgent are better performance, more versatility as a multi-purpose data mover (protect VMs and physical data), and resiliency. If using a tape library, presenting it to a virtualized MediaAgent adds an additional layer of complexity for configuration and troubleshooting. A MediaAgent can be virtualized if all performance requirements including CPU, RAM, index directory location and deduplication database location are being met.

Scenario: Remote Site MediaAgents

You need to protect a smaller remote site and want to keep a local copy of data for quick restore. However, you are concerned about hardware costs for a MediaAgent.

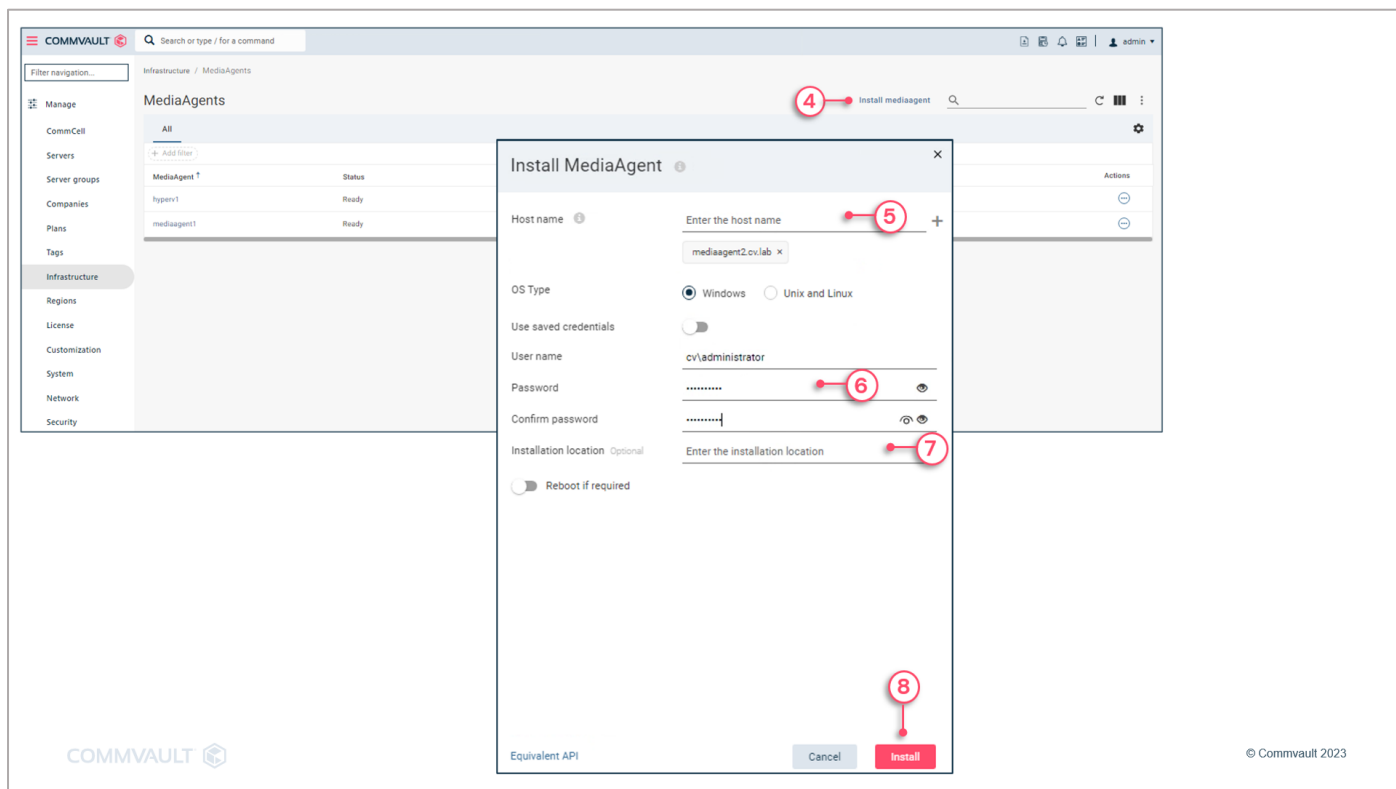
Solution: Virtualize the remote site MediaAgent and keep a shorter retention for the local copy, producing a smaller footprint. Then replicate the data using Secondary, Auxiliary, or DASH Copy to the main data center physical MediaAgent where it can be kept for a longer retention.

Deploy the MediaAgent



1. Select **Manage**
2. Select **Infrastructure**
3. Click the **MediaAgents** tile

Deploy the MediaAgent ...cont'd



4. This window displays all configured MediaAgents; click **Install MediaAgent** to add a new server
5. Type the name of the MediaAgent server and click + to add it to the list; repeat for all MediaAgents you need to install
6. Enter credentials with administrative privileges on the systems
7. Enter the installation location
8. Check to reboot the system upon installation if needed and click **Install**

Deploy the MediaAgent ...cont'd

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Search or type / for a command

Filter navigation...

Infrastructure / MediaAgents

Manage

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Plans

Tags

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Regions

License

Customization

System

Network

Security

Developer tools

Workflows

Web console

MediaAgents

All

+ Add filter

MediaAgent ↑	Status	OS	Version
hyperv1	Ready	Windows Server 2022 Standard	11.32.8
mediaagent1	Ready	Windows Server 2022 Standard	11.32.8

Install MediaAgent

To go to the job details, click the link 42.

9

OK

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- 9. Installation progress can be monitored by clicking the *Job ID link*

Access the Index Directory

The screenshot displays the Commvault MediaAgent configuration interface. The left sidebar contains navigation links: Server groups, Companies, Plans, Tags, Infrastructure (highlighted), Regions, License, Customization, System, Network, Security, Developer tools, and Workflows. The main content area is divided into three sections: Security, Index Cache, and Settings. The Index Cache section is highlighted with a red border and contains the following information:

Index Cache	
Index cache path	C:\Program Files\Commvault\ContentStore\IndexCache
Enable logs cache	☐
Index Directory Space	
Total Size	99.46 GB
Index	224.00 MB
Other data	18.32 GB
Free space	80.92 GB

The right sidebar contains the Settings section, which includes a table of configuration options:

Settings	
Parallel data transfer operations	100
Read ahead count for Performance	256
Optimize Network performance by parallelizing DDB lookup	1
Cloud Storage Server Certificate identity check	1

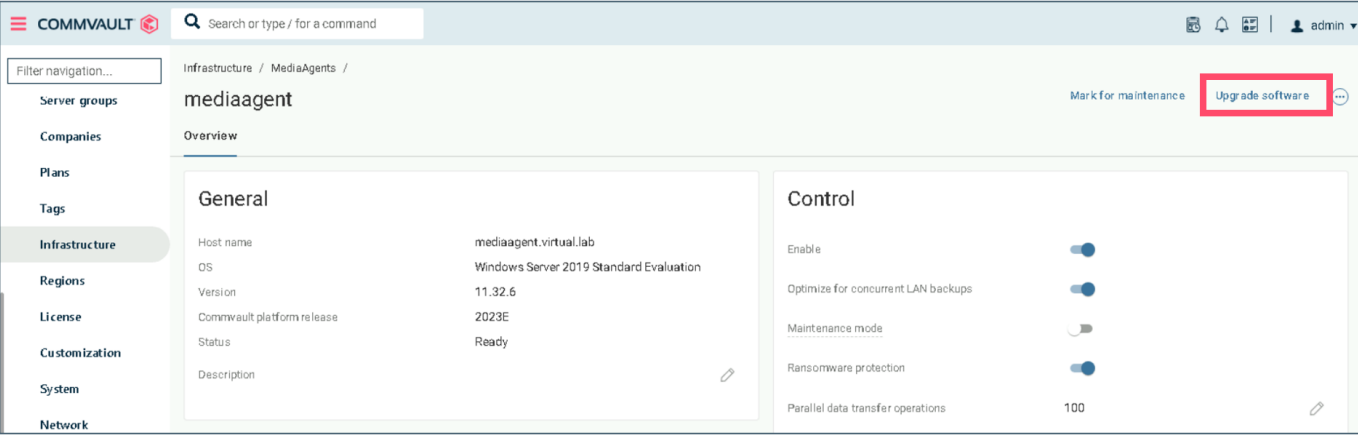
Below the Settings section is the Power management section, which states: "Power management is not enabled on this MediaAgent".

MediaAgent Advanced Options

The index directory must be located on a dedicated high-speed disk, preferably solid-state drives. Index directory performance is critical when streaming a high number of jobs to a MediaAgent and when conducting DASH full operations. When MediaAgent software is first deployed to a server, the location of the index directory will be on the system drive. It is recommended to change the location to a dedicated drive prior to any jobs running.

To change the Index Cache location you can click the **Edit icon** from within the MediaAgent configuration page.

Apply Updates to the MediaAgent



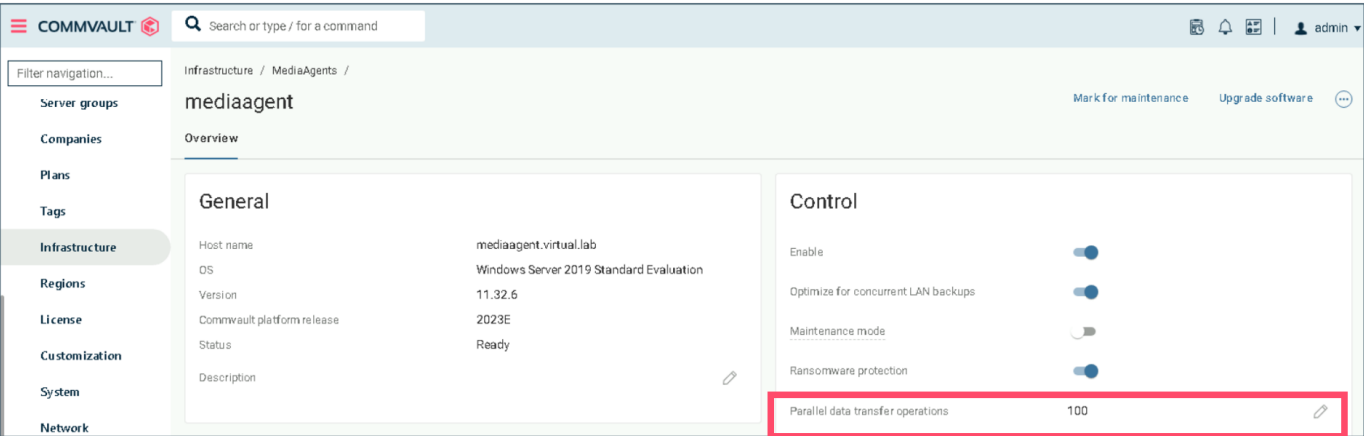
The screenshot displays the Commvault web interface for managing MediaAgents. The left sidebar contains a navigation menu with categories like Server groups, Companies, Plans, Tags, Infrastructure (selected), Regions, License, Customization, System, and Network. The main content area shows the 'mediaagent' overview page. At the top right of this page, there are two buttons: 'Mark for maintenance' and 'Upgrade software', with the latter being highlighted by a red rectangle. Below these buttons, the page is divided into two panels: 'General' and 'Control'. The 'General' panel lists attributes such as Host name (mediaagent.virtual.lab), OS (Windows Server 2019 Standard Evaluation), Version (11.32.6), Commvault platform release (2023E), Status (Ready), and a Description field. The 'Control' panel contains several toggle switches for 'Enable', 'Optimize for concurrent LAN backups', 'Maintenance mode', and 'Ransomware protection', along with a numeric value for 'Parallel data transfer operations' set to 100.

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Software updates can be applied to a MediaAgent directly from the MediaAgent overview page. Click **Upgrade Software** to perform the software update.

Configure Parallel Data Transfer Options



The screenshot displays the Commvault web interface for configuring a MediaAgent. The left sidebar shows navigation options: Server groups, Companies, Plans, Tags, Infrastructure (selected), Regions, License, Customization, System, and Network. The main content area is titled 'mediaagent' and includes an 'Overview' tab. The 'General' section lists details such as Host name (mediaagent.virtual.lab), OS (Windows Server 2019 Standard Evaluation), Version (11.32.6), Commvault platform release (2023E), Status (Ready), and a Description field. The 'Control' section contains several toggle switches: Enable, Optimize for concurrent LAN backups, Maintenance mode, and Ransomware protection. At the bottom of the Control section, the 'Parallel data transfer operations' setting is highlighted with a red box, showing a value of 100 and an edit icon.

Setting	Value
Parallel data transfer operations	100

A MediaAgent can receive and handle a certain amount of parallel data transfer operations, or streams. This is based on hardware resources of the MediaAgent, as well as the Commvault Online Documentation. By default, a MediaAgent has 100 set as a maximum of streams it can receive. After proper evaluation, this number can be increased or decreased.

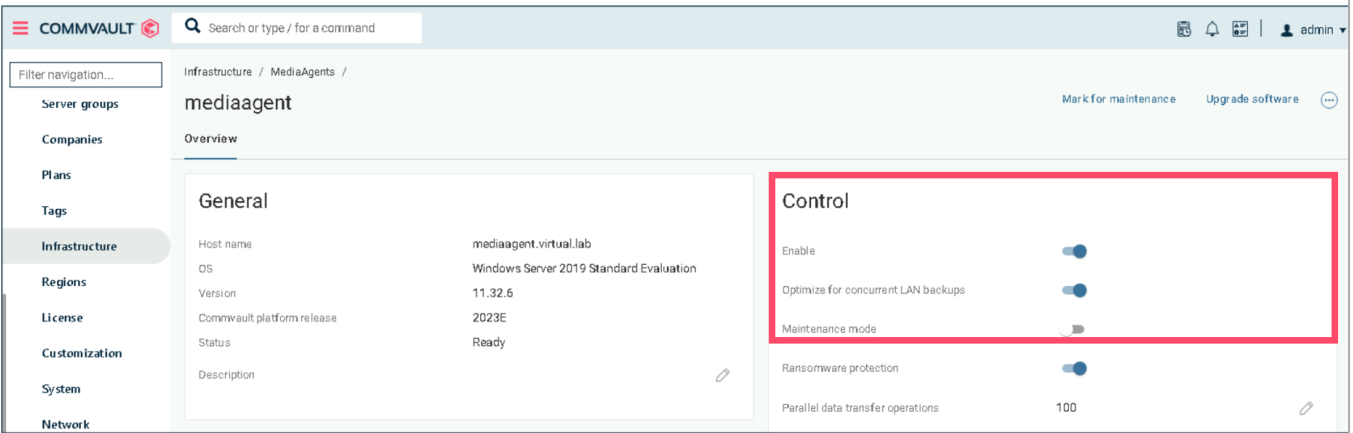
To adjust this value, from the MediaAgent configuration page, click the edit icon next to Parallel Data transfer operations.

Disable MediaAgent vs. Mark MediaAgent Offline

Option	Result
Disable a MediaAgent	No client nor administrative tasks can be executed. The MediaAgent is unavailable and unusable by the CommCell console.
Mark a MediaAgent offline for maintenance	Temporarily prohibits data protection, data recovery, and auxiliary copy jobs using the MediaAgent. However, administrative tasks such as 'Clean Drives,' 'full scan,' or 'disk library pruning' may still be conducted.

If maintenance is required on the MediaAgent or hardware, it is possible to disable or mark a MediaAgent offline for maintenance. It is important to understand the differences between disabling a MediaAgent and marking it offline for maintenance.

Disable and Offline Settings



The screenshot displays the Commvault web interface for configuring a MediaAgent. The left sidebar shows navigation options: Server groups, Companies, Plans, Tags, Infrastructure (selected), Regions, License, Customization, System, and Network. The main content area is titled 'mediaagent' and includes an 'Overview' tab. The 'General' tab is active, showing details like Host name (mediaagent.virtual.lab), OS (Windows Server 2019 Standard Evaluation), Version (11.32.6), and Status (Ready). The 'Control' pane, highlighted with a red border, contains settings for Enable (toggle), Optimize for concurrent LAN backups (toggle), Maintenance mode (toggle), Ransomware protection (toggle), and Parallel data transfer operations (set to 100). The top navigation bar includes a search bar and user information (admin). The bottom of the page shows the Commvault logo and copyright notice (© Commvault 2023).

You can disable or enable a MediaAgent or mark it offline for maintenance from the Control pane of the MediaAgent configuration page.